PLANCK WINDOW ONTO PARALLEL UNIVERSES

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Abstract. Where could a parallel universe show itself in our sky, and how bright must its mark be? We measure this window on the full Planck 2018 sky. A bubble-collision disc brighter than 153–211 μK, or a cosmic texture brighter than 244–383 μK, would already be seen, and none is. The Cold Spot lies just inside the window, 13 per cent below the detection edge, where a collision disc fits it at two sigma. Polarisation maps can test it.

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1. Introduction and observational motivation

1.1. The cosmic microwave background

The cosmic microwave background (CMB) is the thermal radiation released about 380 000 years after the Big Bang, when the primordial plasma became neutral and photons decoupled from matter. Today it is a blackbody of temperature

$$T_0 = 2.72548 \pm 0.00057\,\mathrm{K} \tag{1.1}$$

(Fixsen, 2009), isotropic to one part in 10^5. The anisotropies are expanded in spherical harmonics,

$$\Delta T(\hat{\mathbf{n}}) = \sum_{\ell \geq 0} \sum_{m=-\ell}^{\ell} a_{\ell m}\, Y_{\ell m}(\hat{\mathbf{n}}), \qquad \hat{C}_\ell = \frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2, \tag{1.2}$$

and their angular power spectrum, measured by WMAP (Hinshaw et al., 2013) and Planck (Planck Collaboration, 2020a,c), is fitted by the six-parameter ΛCDM model over three decades in multipole (Figure 2). For a statistically isotropic Gaussian sky the $a_{\ell m}$ are independent with variance C_ℓ, and any feature in a map can be compared with what such a sky produces by chance.

1.2. Anomalies

A few large-angle features persist in both WMAP and Planck data at the level of two to three standard deviations (Schwarz et al., 2016; Planck Collaboration, 2020d). Two motivated the earlier versions of this paper.

The Cold Spot. Found by Vielva et al. (2004) with a spherical Mexican-hat wavelet, the Cold Spot is a depression of about $-70\,\mu$K at the resolution of the original analysis, centred at Galactic coordinates $(l, b) \approx (209°, -57°)$, about 5° in radius and surrounded by a warmer ring (Cruz et al., 2005; Vielva, 2010). A Gaussian sky produces a wavelet feature this extreme in roughly one to two per cent of realisations, depending on how the freedom of choosing scale and position is counted (Cruz et al., 2005; Planck Collaboration, 2020d). Figure 1 shows it in the Planck map.[1]

The axis of evil. The quadrupole and octupole of the CMB are aligned with each other and roughly with the ecliptic (de Oliveira-Costa et al., 2004; Schwarz et al., 2016).

1.3. Explanations proposed for the Cold Spot

The proposals fall into four groups:

(i) a foreground void whose integrated Sachs–Wolfe effect cools the photons that cross it (Inoue & Silk, 2006; Szapudi et al., 2015; Nadathur et al., 2014; Kovács et al., 2022);
(ii) a collapsing cosmic texture, a topological defect of a broken global symmetry (Turok & Spergel, 1990; Cruz et al., 2007; Vielva, 2010);
(iii) a collision between our bubble universe and a neighbouring one in eternal inflation (Chang et al., 2009; Aguirre & Johnson, 2011; Feeney et al., 2011);
(iv) contact with a conjugate parallel universe, proposed in v1 of this paper and developed in Part I.

Proposals (ii)–(iv) all leave an azimuthally symmetric spot on the last-scattering surface, and Part I shows that (iv) leaves one of exactly the shapes that (ii) and (iii) leave. This is what makes the conjugate-universe hypothesis testable.

[1] Versions 1 and 2 identified the Cold Spot with the Eridanus supervoid at $z \approx 1$. The two are different objects. The Cold Spot is a feature of the last-scattering surface at $z \approx 1100$. The Eridanus supervoid is a large underdensity of galaxies at $z \approx 0.2$ along the same line of sight (Szapudi et al., 2015; Kovács et al., 2022); in ΛCDM its integrated Sachs–Wolfe imprint accounts for only part of the Cold Spot (Nadathur et al., 2014; Mackenzie et al., 2017).

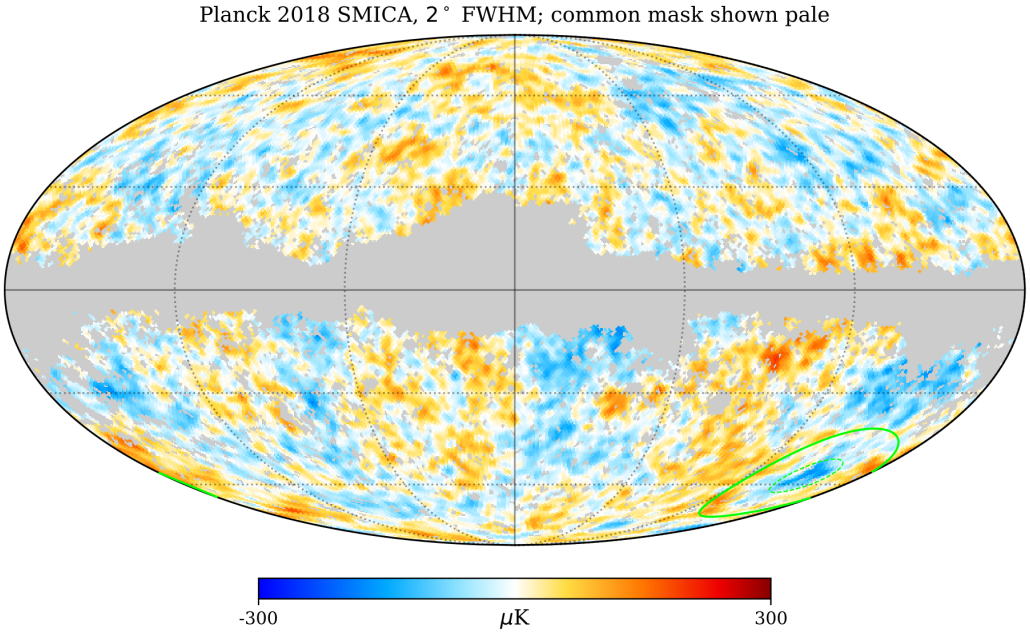


Figure 1. The Planck 2018 SMICA map at 2° resolution in Galactic Mollweide projection; masked pixels are pale. The circles mark the Cold Spot at radii 6° (dashed) and 16° (solid).

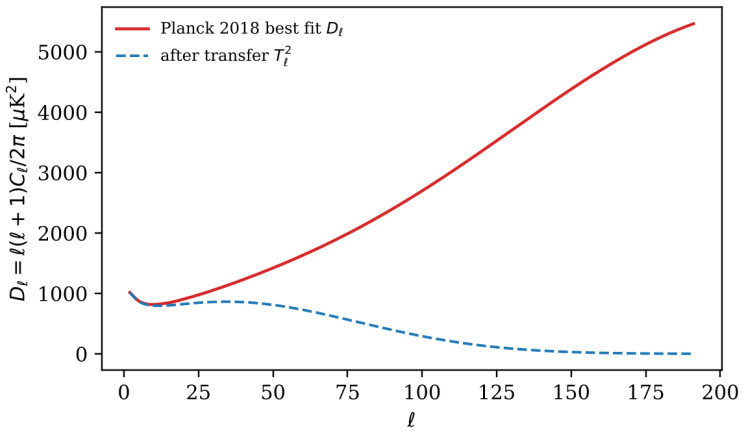


Figure 2. The Planck 2018 best-fit spectrum $D_\ell = \ell(\ell+1)C_\ell/2\pi$ (solid) and the same spectrum after the instrumental beam, pixel window and 2° smoothing used in Part II (dashed).

1.4. What this version does

Versions 1 and 2 argued that the Cold Spot is the imprint of a parallel universe. An argument of that kind can be settled only by data. Version 3 therefore asks a measurable question: *which imprints of a parallel universe does the Planck sky still allow?* We call the set of allowed amplitudes and sizes the *observable window*. Part I restates the theoretical framework and derives the imprint. Part II measures the window with the four Planck 2018 temperature maps, by a targeted fit at the Cold Spot and by a blind search of the whole sky, both calibrated with simulated Gaussian skies.

The results are these. At the Cold Spot a collision disc of radius 16° improves the fit by $\Delta\chi^2 = 23.1$; Gaussian skies that repeat the selection of the coldest spot do as well with probability 0.050, a two-sigma preference (Section 12). Over the whole sky no collision and no texture is significant; the global p-values lie between 0.31 and 0.67 (Section 13). Injected signals turn this into the bounds $\bar{N} < 4.7$ for collisions with central temperature $|\Delta T_0| \geq 210\,\mu$K and $\bar{N} < 4.6$

for textures with $|\Delta T_0| \geq 350\,\mu$K at 95 per cent confidence (Section 14). The texture fit to the Cold Spot, $-215 \pm 50\,\mu$K at radius 6°, lies 13 per cent below the amplitude of $246\,\mu$K at which the blind search would have flagged it, inside the open window (Section 15). The data neither detect a parallel universe nor exclude the Cold Spot as its imprint.

1.5. Outline

Part I: Section 2 introduces two-dimensional complex time, Section 3 the contour integrals and the affine parameter, Section 4 the entanglement of the universe pair, Section 5 the topological split, Section 6 the junction geometry, Section 7 eternal inflation, Section 8 the holomorphic structure of time and Section 9 textures. Section 10 derives the observable imprint and defines the window. Part II: Section 11 data and null model, Section 12 the targeted test, Section 13 the blind search, Section 14 efficiencies and bounds, Section 15 the window. Section 16 discusses tests and limitations and Section 17 concludes. Appendices A–C give derivations.

Part I

The conjugate-universe framework

2. Two-dimensional complex time

2.1. From one to two temporal dimensions

General relativity describes spacetime as a four-dimensional Lorentzian manifold with one time dimension. The no-boundary proposal of Hartle & Hawking (1983) uses Euclidean time, $t \to -i\tau$, to define the path integral over geometries near the initial singularity. We take the imaginary direction of time as a hidden degree of freedom rather than a computational device, and assume that it is suppressed in the present epoch by a symmetry-breaking transition (Section 8).

Definition 2.1: Two-dimensional complex time

The temporal coordinate is the complex variable

$$T = \tau + i\tilde{\tau}, \tag{2.1}$$

where $\tau \in \mathbb{R}$ is the time measured in the $3+1$-dimensional universe and $\tilde{\tau} \in \mathbb{R}$ is a hidden imaginary time. Locally the time sector is $\mathbb{C} \cong \mathbb{R}^2$.

In this picture time is a closed curve in $\mathbb{R}^2$ rather than a line in $\mathbb{R}$, much as an extra spatial dimension is compactified in Kaluza–Klein theories.

2.2. Two universes with opposite time loops

We model two conjugate universes α and β as three-dimensional Riemannian manifolds $\mathcal{M}_\alpha$, $\mathcal{M}_\beta$, each carrying a closed time curve in its boundary: $C^1 \subset \partial\mathcal{M}_\alpha$ traversed counterclockwise and $C^2 \subset \partial\mathcal{M}_\beta$ traversed clockwise. Their parametrisations are

$$\alpha = \left(e^{it}\right)_1 = (\cos t + i \sin t)_1, \tag{2.2}$$

$$\beta = \left(e^{-it}\right)_2 = (\cos t - i \sin t)_2. \tag{2.3}$$

The real part $\cos t$ is common to both universes; the imaginary part runs forward in α and backward in β.[2] The two curves meet at a junction point J (Figure 3, left), and their union $C^1 \cup_J C^2$ is a closed contour of imaginary time. For a family of universes the same construction gives a contour $\Gamma = \sum_n \lambda_n$ of closed curves λ_n, each with identified endpoints.

2.3. Consequences of opposite orientations

The first consequence concerns the arrow of time. In universe α the imaginary part grows with t, which we identify with the direction of increasing entropy; in β it decreases. The second concerns the vacuum. The imaginary direction governs the analytic continuation of quantum field theory to Euclidean signature. If the two universes have opposite imaginary orientations, vacuum fluctuations at the junction carry opposite phases. In the framework this is the mechanism by which the junction leaves a cold or hot imprint; Section 10 derives its shape.

The curves C^1 and C^2 have codimension one in the three-dimensional spatial section. A measurement confined to that section does not resolve the hidden direction, just as extra compact spatial dimensions are invisible to low-energy observers.

[2] This is related to, but distinct from, the CPT-symmetric universe of Boyle et al. (2018), in which the universe before the Big Bang is the CPT image of the universe after it.

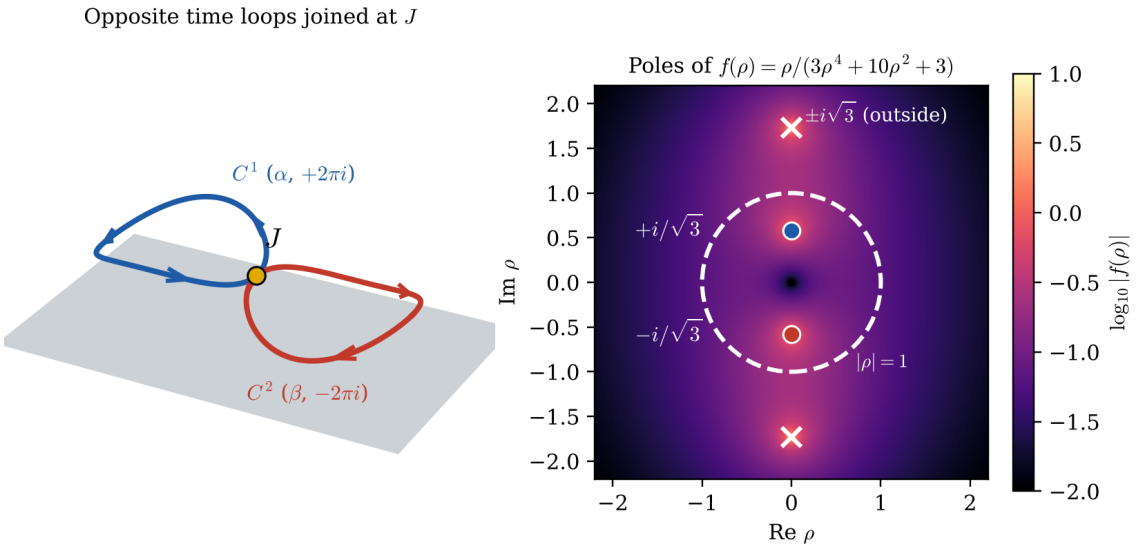


Figure 3. Left: the two time loops, C^1 of universe α (counterclockwise, winding $+2\pi i$) and C^2 of universe β (clockwise, $-2\pi i$), joined at the junction J; the vertical offset indicates the hidden imaginary direction. Right: modulus of the integrand of the affine-parameter integral (3.3) in the ρ plane. Two poles lie inside the unit circle (dashed), two outside.

3. Contour integrals and the affine parameter

3.1. Winding numbers

With $\rho(t) = e^{it}$, the integral of $1/\rho$ around C^1 is

$$\oint_{C^1} \frac{d\rho}{\rho} = \int_0^{2\pi} \frac{ie^{it}}{e^{it}}\, dt = 2\pi i, \tag{3.1}$$

and around the clockwise C^2 it is $-2\pi i$. These are the winding numbers that distinguish the two universes.

3.2. The affine parameter

To obtain a real parameter for the coupled pair we evaluate

$$I = \int_{-\pi}^{\pi} \frac{dt}{1 + 3\cos^2 t}. \tag{3.2}$$

With $\rho = e^{it}$, $\cos t = \tfrac{1}{2}(\rho + \rho^{-1})$ and $dt = d\rho/(i\rho)$,

$$1 + \tfrac{3}{4}(\rho + \rho^{-1})^2 = \frac{3\rho^4 + 10\rho^2 + 3}{4\rho^2}, \qquad I = -4i \oint_{|\rho|=1} f(\rho)\, d\rho, \quad f(\rho) = \frac{\rho}{3\rho^4 + 10\rho^2 + 3}. \tag{3.3}$$

The denominator factorises as $3(\rho^2 + 3)(\rho^2 + \tfrac{1}{3})$, with simple poles at $\pm i\sqrt{3}$ outside the unit circle and $\pm i/\sqrt{3}$ inside it (Figure 3, right).

Theorem 3.1: Affine parameter

$\mathrm{Res}(f, i/\sqrt{3}) = \mathrm{Res}(f, -i/\sqrt{3}) = \frac{1}{16}$, and

$$I = -4i \cdot 2\pi i \left[\mathrm{Res}(f, \tfrac{i}{\sqrt{3}}) + \mathrm{Res}(f, -\tfrac{i}{\sqrt{3}})\right] = \pi. \tag{3.4}$$

Proof. At a simple zero ρ_0 of the denominator $q(\rho) = 3\rho^4 + 10\rho^2 + 3$, $\mathrm{Res}(f, \rho_0) = \rho_0/q'(\rho_0) = \rho_0/(12\rho_0^3 + 20\rho_0) = 1/(12\rho_0^2 + 20)$. For $\rho_0 = \pm i/\sqrt{3}$, $\rho_0^2 = -\tfrac{1}{3}$, so the residue is $1/(20 - 4) =$

$\frac{1}{16}$. The residue theorem gives $I = -4i \cdot 2\pi i \cdot \frac{1}{8} = \pi$. As a check, the standard integral $\int_0^{2\pi} dt/(a + b\cos^2 t) = 2\pi/\sqrt{a(a+b)}$ with $a = 1$, $b = 3$ gives $2\pi/2 = \pi$. □

In the framework each of the two interior poles is assigned to one of the loops, $+i/\sqrt{3}$ to C^1 and $-i/\sqrt{3}$ to C^2; each contributes $\pi/2$, and the value π serves as the affine parameter of the coupled system, the angle subtended by each loop when the two are merged into one contour.

3.3. The vanishing loop

If the loops have radius R, the length $2\pi R \to 0$ as $R \to 0^+$. We identify this limit with $t = 0$, when the two universes occupied the same infinitesimal volume. Expansion increases R and separates the curves; the junction is the last point of contact.

4. Entanglement of the universe pair

4.1. The landscape

In eternal inflation the string landscape is estimated to contain of order 10^{500} metastable vacua (Bousso & Polchinski, 2000). We consider one pair of universes born together in this landscape and model each as a quantum system, with Hilbert spaces $\mathcal{H}_\alpha$ and $\mathcal{H}_\beta$.

4.2. The state of the pair

The joint state before separation is a pure state in $\mathcal{H}_\alpha \otimes \mathcal{H}_\beta$,

$$|\Psi\rangle_{\alpha\beta} = \sum_{i,j} c_{ij}\, |i\rangle_\alpha \otimes |j\rangle_\beta, \qquad \sum_{i,j} |c_{ij}|^2 = 1. \tag{4.1}$$

Labelling the two orientations $+2\pi i$ and $-2\pi i$ by $|0\rangle$ and $|1\rangle$, the anti-correlated state of maximal entanglement is the singlet

$$|\Psi\rangle_{\alpha\beta} = \tfrac{1}{\sqrt{2}}(|0\rangle_\alpha|1\rangle_\beta - |1\rangle_\alpha|0\rangle_\beta). \tag{4.2}$$

Each universe is then in the opposite orientation of the other, and neither has a definite orientation on its own.

4.3. Reduced state and entropy

The state of universe α alone is the reduced density matrix $\varrho_\alpha = \mathrm{Tr}_\beta\, |\Psi\rangle\langle\Psi|$, with entries $(\varrho_\alpha)_{ii'} = \sum_j c_{ij} c^*_{i'j}$. Its von Neumann entropy

$$S(\varrho_\alpha) = -\,\mathrm{Tr}(\varrho_\alpha \log_2 \varrho_\alpha) = -\sum_k \lambda_k \log_2 \lambda_k, \tag{4.3}$$

where λ_k are the eigenvalues of ϱ_α, measures the entanglement between the two universes. It is zero for a product state and reaches its maximum $\log_2 n$ when $\varrho_\alpha = \mathbb{1}/n$. For the singlet (4.2), $\varrho_\alpha = \mathbb{1}/2$ and $S = 1$ ebit (Appendix A).

4.4. Time evolution

The joint state evolves by $U(t) = \exp(-iHt/\hbar)$. Unitary evolution of the pair leaves the entropy of the joint state unchanged, $S(U\varrho U^\dagger) = S(\varrho) = 0$, but it can change the entropy of each member. For the family $|\Psi(\vartheta)\rangle = \cos\vartheta|01\rangle - \sin\vartheta|10\rangle$,

$$S(\varrho_\alpha) = -\cos^2\vartheta \log_2 \cos^2\vartheta - \sin^2\vartheta \log_2 \sin^2\vartheta, \tag{4.4}$$

which rises from 0 at $\vartheta = 0$ to 1 ebit at $\vartheta = \pi/4$ (Figure 4). If the interaction at the junction drives ϑ towards $\pi/4$ as the universes separate, each universe ends maximally entangled with the other while the pair remains in a pure state. The entanglement is a correlation, not a channel: by the no-signalling theorem no local operation in β changes ϱ_α, so the entanglement itself has no observable consequence in our universe. What can be observed is the classical imprint of the junction (Section 10).

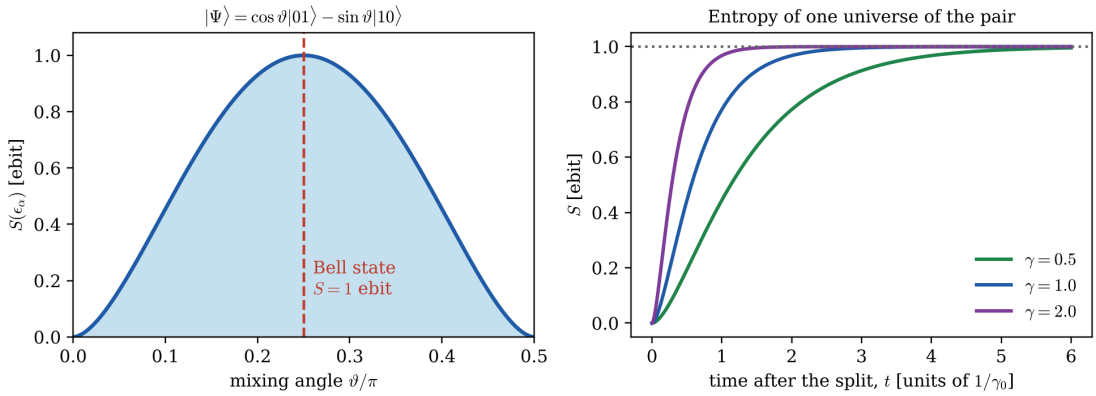


Figure 4. Left: entanglement entropy of one universe for the family $\cos\vartheta|01\rangle - \sin\vartheta|10\rangle$; the singlet ($\vartheta = \pi/4$) carries one ebit. Right: illustration of entropy growth when the mixing angle relaxes as $\vartheta(t) = \frac{\pi}{4}(1 - e^{-\gamma t})$ for three rates γ.

5. Ricci flow and the topological split

5.1. Ricci flow with surgery

How one parent manifold becomes two can be described with the Ricci flow

$$\partial_t g_{ij} = -2R_{ij} \tag{5.1}$$

introduced by Hamilton. On a closed three-manifold the flow can develop finite-time singularities in which a region shaped like $S^2 \times (-1, 1)$ shrinks to a point, a *neck pinch* (Hamilton, 1995). Perelman's completion of Hamilton's programme, presented in full in the monograph of Morgan & Tian (2007), shows that the flow can be continued through such singularities by surgery: the neck is cut out, each side is capped with a ball, and the flow resumes.

5.2. Blow-up and the neck

Let $(M, g(t))$ be a Ricci flow on $[0, T)$ with $T < \infty$ and curvature unbounded as $t \to T$. Choose points (p_i, t_i) with $K_i = |\mathrm{Rm}|(p_i, t_i) \to \infty$. The rescaled flows

$$g_i(s) = K_i\, g(t_i + s/K_i) \tag{5.2}$$

have curvature one at $(p_i, 0)$. Under the bounds supplied by Perelman's non-collapsing theorem they subconverge, by Hamilton's compactness theorem, to an ancient solution with non-negative curvature (Morgan & Tian, 2007). At a neck pinch the limit is the shrinking round cylinder $\mathbb{R} \times S^2$, whose sphere radius obeys $r^2(t) = 2(T - t)$ (Appendix A).

5.3. Twin universes

In the framework the surgery is the birth of the pair. Before the singular time the parent universe M develops a neck; at the singular time the neck is removed and capped, leaving two components M_1 and M_2, our universe and its conjugate (Figure 5). If each component is simply connected and carries positive curvature after the flow, it is diffeomorphic to S^3 (Morgan & Tian, 2007).

The two children are of equal dimension. In linear-algebra terms, if the parent space decomposes as $V = U \oplus Q$, then $V/Q \cong U$ and $V/U \cong Q$, and

$$\operatorname{codim} Q = \dim(V/Q) = \dim V - \dim Q, \qquad \operatorname{codim} U = \dim V - \dim U. \tag{5.3}$$

Equal codimension, $\operatorname{codim} Q = \operatorname{codim} U$, is equivalent to $\dim Q = \dim U$: the two quotients are isomorphic, which is the sense in which the universes are twins. The junction J is the site of the removed neck.

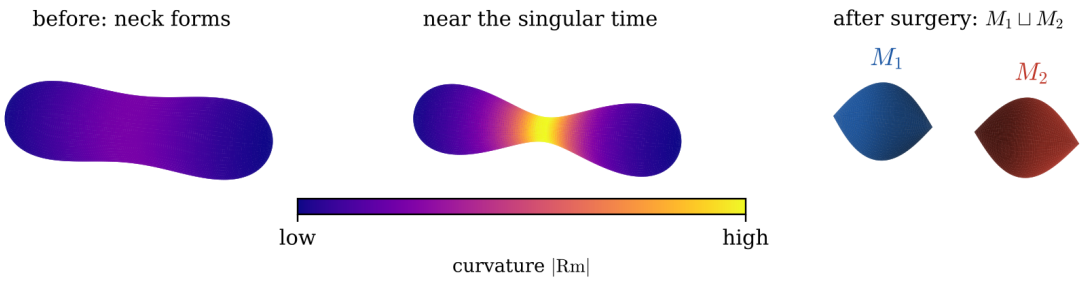


Figure 5. Neck pinch and surgery. Left and centre: a dumbbell-shaped manifold under Ricci flow, coloured by curvature; the neck thins and its curvature grows. Right: after surgery the neck is replaced by two caps, leaving M_1 (our universe) and M_2 (the conjugate universe).

6. Geodesics and the junction

6.1. Geodesics on the time surface

Model the two-dimensional time surface near J as a sphere of constant curvature $K = k^2 > 0$. The Riemann tensor

$$R^k{}_{ij\ell} = \partial_i \Gamma^k_{j\ell} - \partial_\ell \Gamma^k_{ij} + \Gamma^m_{j\ell}\Gamma^k_{im} - \Gamma^m_{ij}\Gamma^k_{\ell m} \tag{6.1}$$

then takes the form $R_{kij\ell} = K(g_{ki}g_{j\ell} - g_{k\ell}g_{ji})$, with Ricci scalar $2K$ and Kretschmann scalar $4K^2$. Geodesics satisfy

$$\frac{d^2\gamma^\lambda}{dt^2} + \Gamma^\lambda_{\mu\nu}\frac{d\gamma^\mu}{dt}\frac{d\gamma^\nu}{dt} = 0, \tag{6.2}$$

and on the sphere they are the great circles. The time curves C^1 and C^2 are modelled as two such geodesics crossing at J.

6.2. Jacobi fields and the six junction configurations

The separation x of nearby geodesics obeys the Jacobi equation

$$\frac{D^2x}{dt^2} + R(x, \dot\gamma)\dot\gamma = 0. \tag{6.3}$$

On a space of constant curvature k^2 every Jacobi field along a unit-speed geodesic is a combination of $\dot\gamma$, $t\dot\gamma$, and $\sin(kt)e(t)$, $\cos(kt)e(t)$ with e a parallel unit normal. The framework labels the possible ways in which the two time curves can meet at J by the endpoint types of these fields: each curve can meet J in an attractive (A) or a separating (S) way, giving four labels A', S' for C^1 and A'', S'' for C^2, and the six pairings of these four labels define six junction configurations $\beta_0, \dots, \beta_5$, the *buffer points*, through which the phase of one curve is matched to the other. Two geodesics leaving the same point at angle τ on the unit sphere are at distance

$$d(t) = \arcsin\Big(\sin t\, \sin\tau\, \sqrt{1 + \cos^2 t\, \tan^2(\tau/2)}\Big) \tag{6.4}$$

at arc length t, as long as $d \le \pi/2$ (Appendix A).

7. Eternal inflation

7.1. Expansion

Galaxies recede according to $v = H_0 D$. Planck gives $H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Planck Collaboration, 2020c); the local distance ladder gives 73.04 ± 1.04 (Riess et al., 2022). The difference, the Hubble tension, is not used in this paper.

7.2. Eternal inflation and bubble universes

Inflation (Guth, 1981) is generically eternal: quantum fluctuations of the inflaton regenerate inflating regions faster than they decay (Vilenkin, 1983; Linde, 1986; Guth, 2007). Most of the volume is always inflating; regions of true vacuum nucleate in it as bubbles, each of which is an infinite open universe to its inhabitants (Garriga et al., 2007; Aguirre & Johnson, 2011). An inflating domain of Hubble volume produces about $e^{3H\Delta t}$ new domains in a time Δt, so the number of domains grows as $\mathcal{N}(t) \sim e^{3Ht}$.

Two scenarios for a pair of universes are distinguished by their later history:

$$f_1(\sigma) = \begin{cases} x = -y, & y < 0 \\ x = y, & y > 0 \end{cases} \qquad f_2(\sigma) = \begin{cases} y = \sin x \\ y = -\sin x \end{cases} \qquad (x > 0). \tag{7.1}$$

In the first the universes separate once and for all, leaving one junction. In the second they meet periodically, at the common zeros of $\pm \sin x$, and each meeting leaves a junction. The second scenario predicts several imprints; Part II tests both through the bound on the number $\bar{N}$ of imprints on the sky.

8. The holomorphic structure of time

8.1. Before and after the transition

The framework models the passage from imaginary to real time as a symmetry-breaking transition $\bar{D} \to \overline{Sp} \to D$, from imaginary time before the Big Bang ($\bar{D}$) through the transition ($\overline{Sp}$) to real time after it (D). The transition is driven by a complex scalar field φ with Lagrangian

$$\mathcal{L} = \partial^\nu \varphi^* \partial_\nu \varphi - V(\varphi), \qquad V(\varphi) = |\varphi|^4 - 10|\varphi|^2. \tag{8.1}$$

The potential has a local maximum at $\varphi = 0$ and a circle of minima at $|\varphi| = \sqrt{5}$, where $V = -25$ (Figure 6): from $dV/d|\varphi|^2 = 2|\varphi|^2 - 10 = 0$. In the false vacuum $\varphi = 0$ both time directions are equivalent; in the true vacuum the phase of φ is fixed and the imaginary direction is hidden.

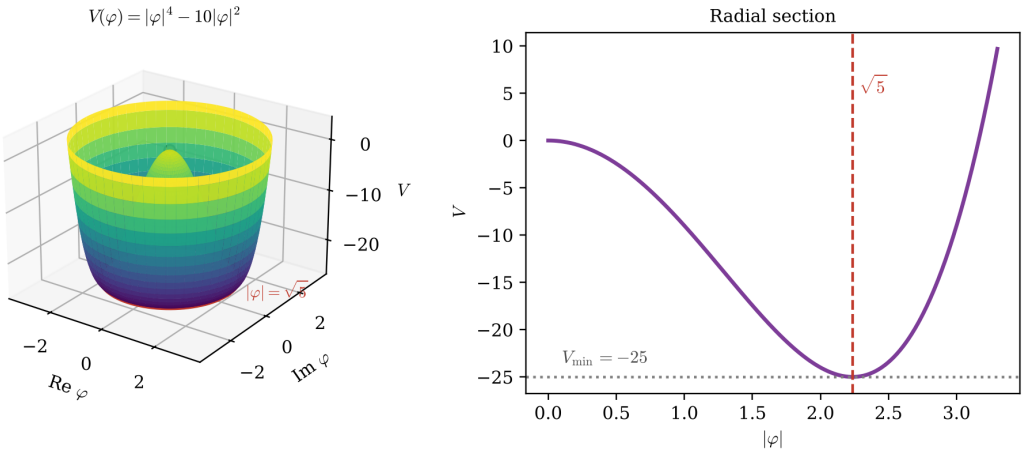


Figure 6. The symmetry-breaking potential (8.1). Left: V over the complex φ plane; the circle of minima at $|\varphi| = \sqrt{5}$ is marked. Right: radial section.

8.2. Holomorphic functions of complex time

A function $f(T) = u + iv$ of complex time is holomorphic when it satisfies the Cauchy–Riemann equations

$$\frac{\partial u}{\partial \tau} = \frac{\partial v}{\partial \tilde{\tau}}, \qquad \frac{\partial u}{\partial \tilde{\tau}} = -\frac{\partial v}{\partial \tau}, \tag{8.2}$$

equivalently $\partial f / \partial \bar{T} = 0$. Then $f'(T) = \partial f/\partial \tau = -i\, \partial f/\partial \tilde{\tau}$: the derivative along real time determines the derivative along the hidden direction. In the framework the physical fields are

holomorphic in T, which is what allows the history in imaginary time to be continued uniquely into real time.

8.3. Causality

In $\bar{D}$ time is purely imaginary and there is no causal order in the Lorentzian sense, as in the no-boundary proposal (Hartle & Hawking, 1983). In D time is real and the usual causal structure holds. The hidden direction persists in D only through the junction, which is a single event in the early universe. Its causal future is therefore a cone, and Section 10 uses exactly this fact.

9. Cosmic textures

9.1. Topological defects

When a symmetry group G breaks to H, the topology of the vacuum manifold $\mathcal{V} = G/H$ decides which defects form (Kibble, 1976): domain walls if $\pi_0(\mathcal{V}) \neq 0$, strings if $\pi_1(\mathcal{V}) \neq 0$, monopoles if $\pi_2(\mathcal{V}) \neq 0$ and textures if $\pi_3(\mathcal{V}) \neq 0$. Textures are unstable; they collapse at the speed of light and leave a hot or cold spot on the CMB (Turok & Spergel, 1990).

Remark 9.1: Which field makes textures

The complex field of (8.1) has vacuum manifold S^1, with $\pi_1(S^1) = \mathbb{Z}$ and $\pi_3(S^1) = 0$. It forms strings, not textures. Versions 1 and 2 used it for both. Textures require, for example, a four-component real field Φ with the same potential $V = |\Phi|^4 - 10|\Phi|^2$; its vacuum manifold is the sphere $|\Phi| = \sqrt{5}$, that is S^3, and $\pi_3(S^3) = \mathbb{Z}$. Everything below assumes this extension.

9.2. The texture profile

A collapsing texture of amplitude ε and scale θ_c produces (Turok & Spergel, 1990; Cruz et al., 2007)

$$\frac{\Delta T}{T_0} = \varepsilon\, g(\theta), \qquad g(\theta) = \begin{cases} \left(1 + 4\theta^2/\theta_c^2\right)^{-1/2}, & \theta \le \theta_*, \\ \tfrac{1}{2}\exp\!\left[-(\theta^2 - \theta_*^2)/(2\theta_c^2)\right], & \theta > \theta_*, \end{cases} \qquad \theta_* = \tfrac{\sqrt{3}}{2}\theta_c. \tag{9.1}$$

Versions 1 and 2 wrote $\theta^2 + \theta_*^2$ in the exponent; with that sign the outer branch starts at $\frac{1}{2}e^{-3/4}$ and the profile jumps at θ_*. With the sign above it is continuously differentiable:

Lemma 9.2: Smooth matching

$g(\theta_*) = \frac{1}{2}$ from both sides, and both branches have slope $-\theta_*/(2\theta_c^2)$ at θ_*.

Proof. At θ_*, $1 + 4\theta_*^2/\theta_c^2 = 4$, so the inner branch equals $\frac{1}{2}$; the outer branch is $\frac{1}{2}e^0 = \frac{1}{2}$. The inner slope is $-\frac{1}{2} \cdot 4^{-3/2} \cdot 8\theta_*/\theta_c^2 = -\theta_*/(2\theta_c^2)$; the outer slope is $\frac{1}{2} \cdot (-\theta_*/\theta_c^2) = -\theta_*/(2\theta_c^2)$. □

9.3. The evidence, then and now

Cruz et al. (2007) fitted (9.1) to the Cold Spot in WMAP data and found $\varepsilon = 7.3^{+2.5}_{-3.6} \times 10^{-5}$ and $\theta_c = 4.9^{+2.8}_{-2.4}$ degrees, with a Bayesian evidence ratio in favour of a texture. Versions 1 and 2 quoted that result. Two later analyses changed the picture. Feeney et al. (2012) searched the whole WMAP sky for textures, accounting for the freedom of choosing the position, and found no evidence for them. Part II repeats the test with Planck: the texture fit to the Cold Spot gives $\varepsilon = (-7.9 \pm 1.8) \times 10^{-5}$ at $\theta_c = 6°$, in agreement with WMAP, but its global significance is $p = 0.55$ (Section 13).

9.4. Textures and the conjugate universe

The two interpretations are compatible. If the transition of Section 8 is driven by the four-component field of Remark 9.1, the junction is the natural site of a texture: the field winds once around the vacuum S^3 where the two universes last touched. The texture amplitude ε then measures the strength of the junction.

10. From the junction to the sky: the observable imprint

The preceding sections describe the conjugate universe in terms that cannot be observed directly: a hidden time direction, an entangled state, a removed neck. This section derives what can be observed.

10.1. The imprint is a disc

The junction is a single event in the early universe (Section 8). Whatever it does to the matter and radiation of our universe propagates no faster than light, so its effect is confined to the future light cone of the event. The same is true of a collision with another bubble universe in eternal inflation (Chang et al., 2009; Aguirre & Johnson, 2011), and the geometry is the same (Figure 7). The observer sees the last-scattering surface as a sphere; the future light cone of the event, propagated to the time of last scattering, is a ball whose boundary cuts that sphere in a circle (Figure 8). The imprint therefore lies inside a disc on the sky.

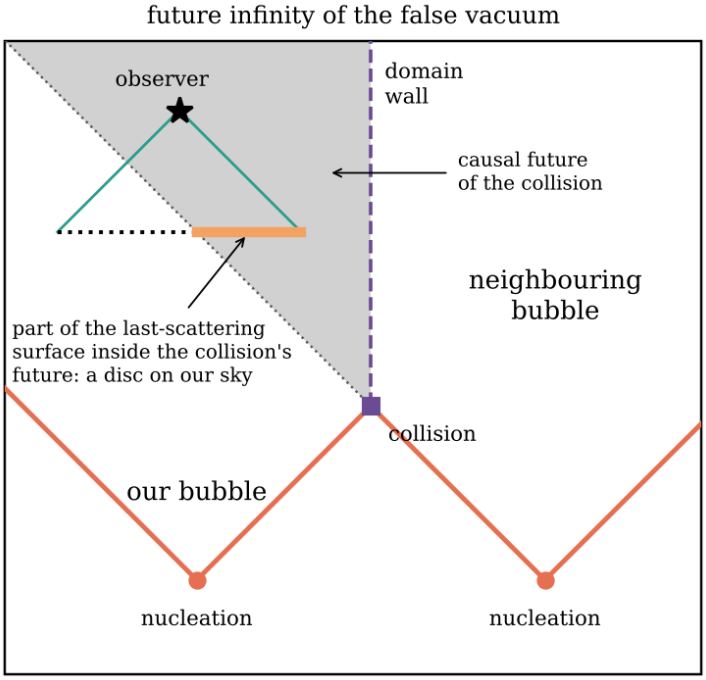


Figure 7. Conformal diagram of the junction event (or of a collision between two bubbles). Walls move on 45° lines; after the event a wall (dashed) separates the two regions. The shaded region is the causal future of the event; the observer's past light cone meets the last-scattering slice partly inside it (thick segment), which on the sky is a disc.

10.2. The profile

The junction singles out one direction $\hat{\mathbf{n}}_0$ in our universe, the direction towards the event, and the configuration is symmetric under rotations about it. To leading order the comoving curvature perturbation is a linear function of distance along $\hat{\mathbf{n}}_0$ on the side of the event and zero on the other (Chang et al., 2009):

$$\zeta(\mathbf{x}) = \lambda\,(\mathbf{x}\cdot\hat{\mathbf{n}}_0 - x_c)\,\Theta(\mathbf{x}\cdot\hat{\mathbf{n}}_0 - x_c), \tag{10.1}$$

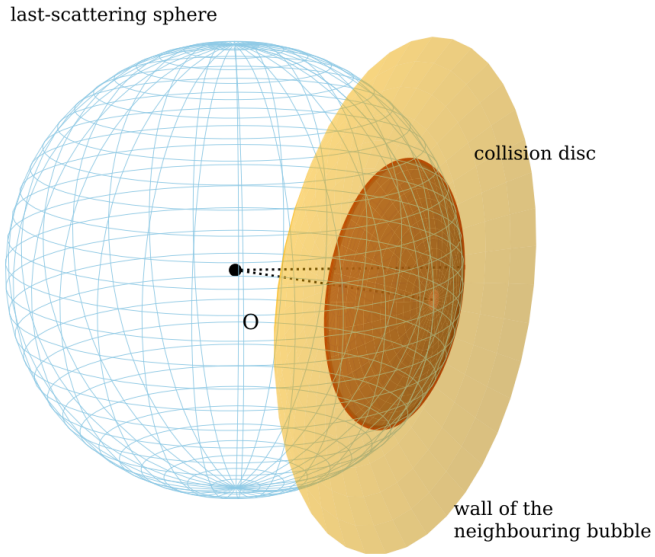


Figure 8. The imprint seen from the observer O at the centre of the last-scattering sphere (wire frame). The boundary of the causal future of the event (light surface) cuts the sphere in a circle; the perturbed region is the cap of angular radius θ_c.

with slope λ, a plane at distance x_c and the Heaviside function Θ. On large scales the Sachs–Wolfe relation gives $\Delta T/T_0 = -\zeta/5$ at the last-scattering radius $r_{\rm ls}$.[3]

Proposition 10.1: Disc profile

Let $\cos\theta = \hat{\mathbf{n}}\cdot\hat{\mathbf{n}}_0$ and $\cos\theta_c = x_c/r_{\rm ls}$ with $0 < x_c < r_{\rm ls}$. Then

(10.2)
$$\frac{\Delta T}{T_0}(\hat{\mathbf{n}}) = \begin{cases} z_0\, w_0(\theta), & \theta < \theta_c, \\ 0, & \theta \geq \theta_c, \end{cases} \qquad w_0(\theta) = \frac{\cos\theta - \cos\theta_c}{1-\cos\theta_c}, \qquad z_0 = -\frac{\lambda r_{\rm ls}}{5}(1-\cos\theta_c).$$

The profile is linear in $\cos\theta$, equals z_0 at the centre and vanishes continuously at the edge.

Proof. At $\mathbf{x} = r_{\rm ls}\hat{\mathbf{n}}$, $\mathbf{x}\cdot\hat{\mathbf{n}}_0 - x_c = r_{\rm ls}(\cos\theta - \cos\theta_c)$, which is positive exactly when $\theta < \theta_c$. Inserting into (10.1) and $\Delta T/T_0 = -\zeta/5$ gives $-\frac{1}{5}\lambda r_{\rm ls}(\cos\theta - \cos\theta_c)$ inside the disc and zero outside, which is (10.2). □

The sign of λ is not fixed, so the centre can be cold or hot; the opposite phases of Section 2 do not decide it. Effects beyond the linear Sachs–Wolfe estimate can add a jump at the edge. Following Feeney et al. (2011) we therefore also use the two-amplitude template

$$\frac{\Delta T}{T_0} = z_0\, w_0(\theta) + z_{\rm crit}\, w_1(\theta) \quad (\theta < \theta_c), \qquad w_1 = 1 - w_0 = \frac{1-\cos\theta}{1-\cos\theta_c}, \tag{10.3}$$

and zero outside. If the junction carries a texture (Section 9), the imprint is instead the texture profile (9.1). Three template families thus cover the imprints the framework allows: the one-amplitude Sachs–Wolfe disc (SW), the two-amplitude disc (B) and the texture (T).

10.3. The observable window

The framework fixes the shape of the imprint but not its amplitude or size, which depend on the unknown strength of the junction and on its distance. The question for the data is therefore which amplitudes and sizes are still allowed.

[3] With the opposite sign convention for ζ the slope λ changes sign; the free amplitude absorbs this.

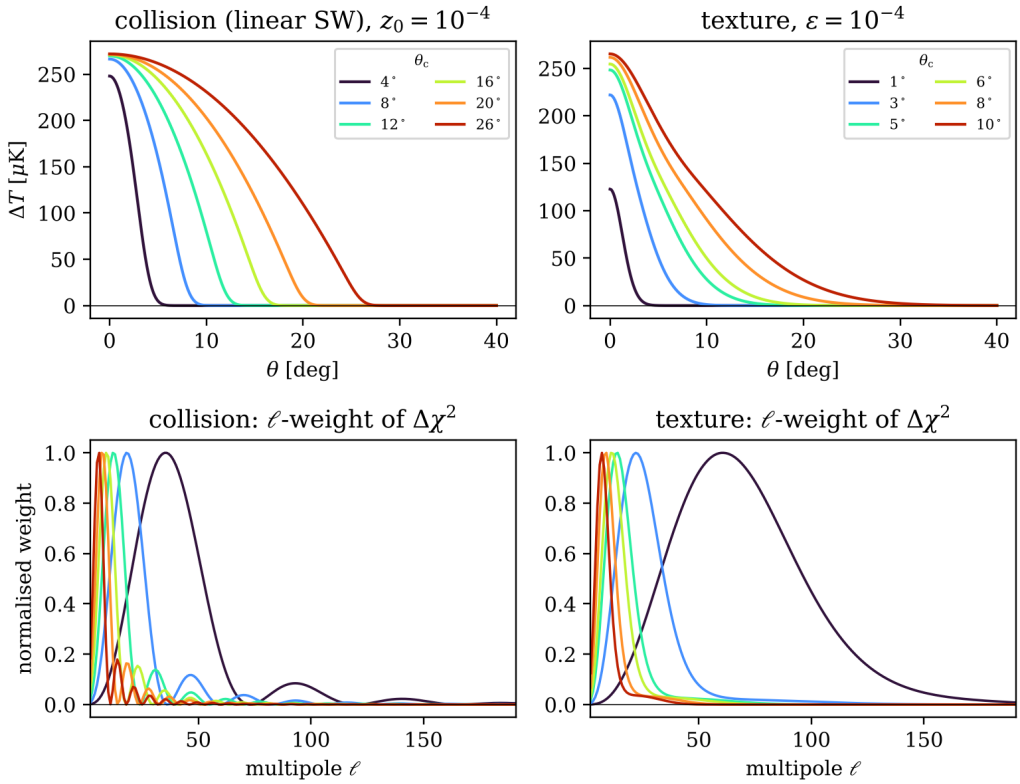


Figure 9. The three template families before (dashed) and after (solid) the 2° smoothing of Part II, for several radii: SW disc (10.2), two-amplitude disc (10.3) (edge component w_1) and texture (9.1).

Definition 10.2: Observable window

For a template family with one amplitude A (central temperature) and radius θ_c, let $\sigma_A(\theta_c)$ be the one-sigma error of A in a fit at the true position and size, and let τ_{95} be the level that the largest $\Delta\chi^2$ over the whole searched sky exceeds with probability 0.05 in Gaussian skies. The *detection edge* is $A_{50}(\theta_c) = \sqrt{\tau_{95}}\,\sigma_A(\theta_c)$ and the *closure line* is $A_{95}(\theta_c) = (\sqrt{\tau_{95}} + 1.645)\,\sigma_A(\theta_c)$. The *open window* is the set of $(\theta_c, |A|)$ with $|A| < A_{50}(\theta_c)$, together with all sizes outside the searched range.

Proposition 10.3: Meaning of the edge

Let an imprint of amplitude A and size θ_c on the searched grid be centred at a position accepted by the search. Then the probability that the search reports a feature above τ_{95} at that position is at least $\Phi(|A|/\sigma_A - \sqrt{\tau_{95}})$, where Φ is the standard normal distribution function. It is at least $\frac{1}{2}$ for $|A| \geq A_{50}$ and at least 0.95 for $|A| \geq A_{95}$.

Proof. At the true position and size the fitted amplitude is $\hat{A} \sim \mathcal{N}(A, \sigma_A^2)$ (Proposition 11.1) and $\Delta\chi^2 = (\hat{A}/\sigma_A)^2$. Hence $P(\Delta\chi^2 > \tau_{95}) = P(|\hat{A}|/\sigma_A > \sqrt{\tau_{95}}) \geq P(\hat{A}\,\mathrm{sgn}A/\sigma_A > \sqrt{\tau_{95}}) = \Phi(|A|/\sigma_A - \sqrt{\tau_{95}})$. The search statistic at that position is the maximum of $\Delta\chi^2$ over sizes, which is at least its value at the true size. At $|A| = A_{50}$ the argument of Φ is 0; at $|A| = A_{95}$ it is 1.645 and $\Phi(1.645) = 0.95$. □

An imprint above the closure line would have been seen with at least 95 per cent probability anywhere in the searched sky, so its absence excludes it. An imprint below the edge would most likely have been missed. The window is the region where the parallel-universe hypothesis is still

alive; Part II measures σ_A, τ_{95} and the efficiency of the full search, and places the Cold Spot in this plane.

Part II

The observable window in the Planck 2018 sky

11. Data and the Gaussian null model

11.1. Maps

We use the four foreground-cleaned temperature maps of the Planck 2018 release, SMICA, NILC, Commander and SEVEM (Planck Collaboration, 2020a,b), at HEALPix resolution $N_{\rm side} = 2048$ (Górski et al., 2005; Zonca et al., 2019), with the common temperature confidence mask. Each map is transformed to spherical harmonics, smoothed with a Gaussian beam of 2° full width at half maximum, truncated at $\ell_{\rm max} = 191$ and synthesised at $N_{\rm side} = 64$ (49 152 pixels of 0.92°). The total transfer function is

$$T_\ell = B_\ell(5')\, W_\ell(2048)\, G_\ell(2^\circ), \qquad G_\ell(\vartheta) = \exp[-\tfrac{1}{2}\ell(\ell+1)\sigma_\vartheta^2], \quad \sigma_\vartheta = \vartheta/\sqrt{8\ln 2}, \tag{11.1}$$

the product of the Planck beam, the pixel window and the smoothing. Monopole and dipole are fitted on the unmasked pixels and removed. After smoothing only 0.6 per cent of the pixel variance comes from $\ell > 120$, so the band limit loses nothing of consequence (Figure 2). All four maps, the mask and the spectrum are public at the Planck Legacy Archive.

11.2. Null model

Under the null hypothesis $\mathfrak{H}_0$ the sky is a statistically isotropic Gaussian field with the Planck 2018 best-fit ΛCDM spectrum C_ℓ (Planck Collaboration, 2020c). After smoothing, the spectrum of the data is $S_\ell = C_\ell T_\ell^2 + N_\ell$, where $N_\ell = \sigma_N^2\, 4\pi/N_{\rm pix}$ is a white-noise term with $\sigma_N = 1\,\mu$K per pixel. The instrumental noise of all four maps is far below this level after smoothing; N_ℓ only keeps the inversions well conditioned. In pixel space the covariance is

$$\mathsf{C}_{ij} = \sum_{\ell=2}^{\ell_{\rm max}} \frac{2\ell+1}{4\pi}\, C_\ell T_\ell^2\, P_\ell(\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j) + \sigma_N^2\, \delta_{ij}, \tag{11.2}$$

which is positive definite (Appendix B).

11.3. Template fits

Let $\mathbf{d}$ be the data in n pixels and $\mathsf{T} \in \mathbb{R}^{k\times n}$ the smoothed templates of one family at one size and centre, scaled so that the amplitudes $\mathbf{a}$ are dimensionless $\Delta T/T_0$ ($k = 2$ for B, $k = 1$ for SW and T). With $\mathsf{A} = \mathsf{T}\mathsf{C}^{-1}\mathsf{T}^\top$ and $\mathbf{b} = \mathsf{T}\mathsf{C}^{-1}\mathbf{d}$ the generalised least-squares (GLS) fit is:

Proposition 11.1: GLS fit

If C is positive definite and T has rank k, then $\chi^2(\mathbf{a}) = \chi^2(\mathbf{0}) - \mathbf{b}^\top\mathsf{A}^{-1}\mathbf{b} + (\mathbf{a}-\hat{\mathbf{a}})^\top\mathsf{A}(\mathbf{a}-\hat{\mathbf{a}})$ with $\hat{\mathbf{a}} = \mathsf{A}^{-1}\mathbf{b}$, so the improvement over the null is $\Delta\chi^2 = \mathbf{b}^\top\mathsf{A}^{-1}\mathbf{b}$. Under the model $\hat{\mathbf{a}}$ is unbiased with covariance A^{-1}; under $\mathfrak{H}_0$, for one fixed template, $\Delta\chi^2$ has the χ^2 distribution with k degrees of freedom.

The proof is in Appendix B. The one-sigma error of an amplitude is the square root of a diagonal entry of A^{-1}. The χ^2_k distribution holds only for a template fixed in advance; the searches below take the maximum over sizes and positions, and their distributions come from simulations.

12. The targeted test at the Cold Spot

12.1. Set-up

The first test asks whether a template centred near the Cold Spot fits the data better than Gaussian fluctuations can by chance. We fit SMICA and NILC in the disc of radius 30° around $(l, b) = (209°, -57°)$, with the mask degraded to $N_{\rm side} = 64$ and thresholded at 0.99, which leaves 2 971 pixels. The two maps differ there by an rms of 0.8 μK (Figure 12). Template centres range over the 135 pixel centres within 6° of the Cold Spot; disc radii are $3°, \dots, 8°$ in steps of 1° and $10°, \dots, 20°$ in steps of 2°, texture scales $1°, \dots, 8°$ and 10°: 1 620 disc and 1 215 texture configurations in all. The fit uses the pixel covariance (11.2) and its Cholesky factor; templates are smoothed exactly through their Legendre coefficients (Proposition B.2). Figure 10 shows the chain and Figures 11 and 13 the region.

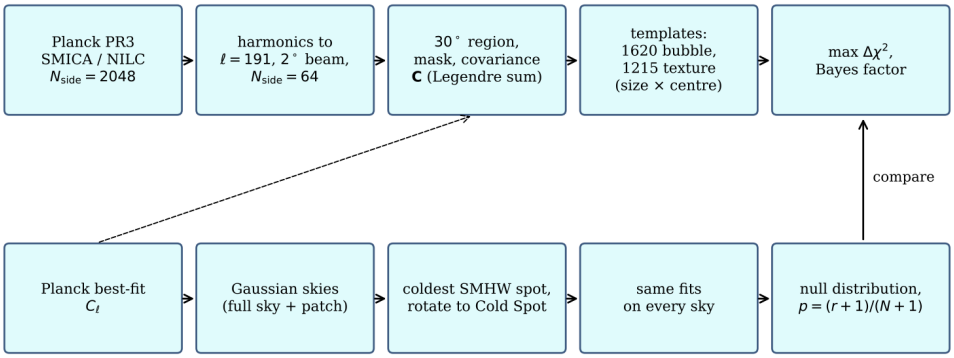


Figure 10. The targeted test. Upper row: the Planck maps are reduced to a 30° region around the Cold Spot and fitted with 1 620 disc and 1 215 texture configurations. Lower row: Gaussian skies with the Planck spectrum pass through the same selection and fits and give the null distributions.

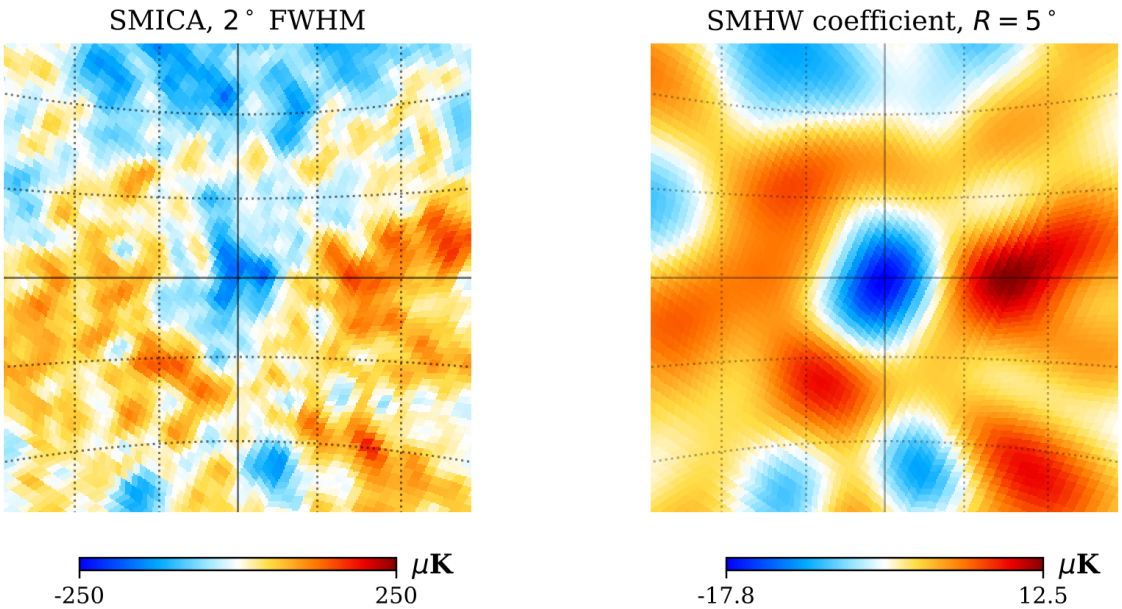


Figure 11. The Cold Spot region in SMICA at 2° resolution (left) and its spherical Mexican-hat wavelet coefficients at scale $R = 5°$ (right). Gnomonic projection centred on $(209°, -57°)$, grid every 10°.

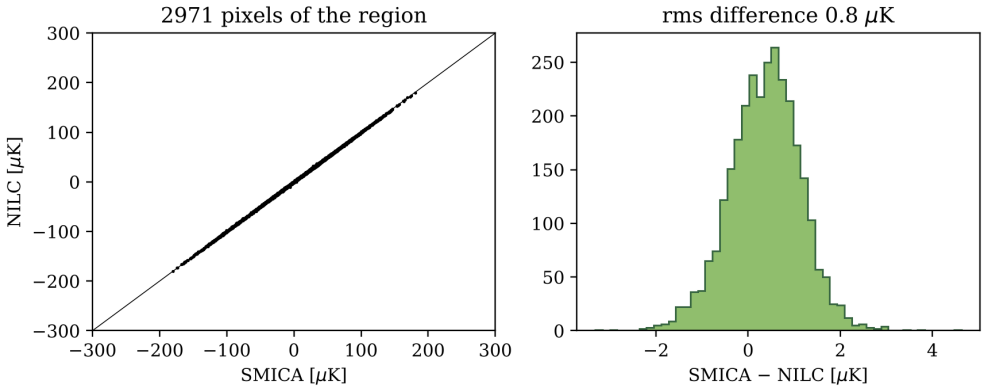


Figure 12. SMICA and NILC in the 2 971 pixels of the targeted region. Left: pixel values against each other. Right: their difference, with an rms of 0.8 μK.

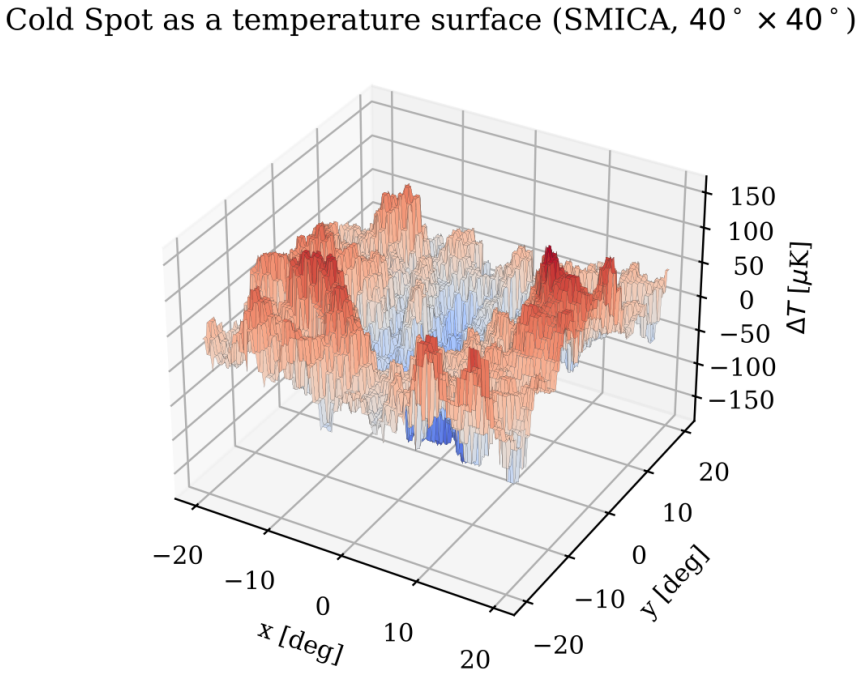


Figure 13. The SMICA temperature in a $40° \times 40°$ square around the Cold Spot drawn as a surface. The Cold Spot is the deep trough near the centre; the warm ring around it is a broken rim.

12.2. Calibration with the selection of the spot

The Cold Spot was singled out because it is extreme, so a fit at its position must be compared with fits at the most extreme position of Gaussian skies. We use two null distributions. At a *fixed position* the null vectors are $\mathsf{L}\mathbf{z}$ with $\mathsf{C} = \mathsf{L}\mathsf{L}^\top$ and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathsf{I})$. For the *selected position* we draw full-sky Gaussian skies, locate the minimum of the spherical Mexican-hat wavelet at $R = 5°$ over the unmasked sky (Vielva et al., 2004), rotate the sky so that this minimum lands on $(209°, -57°)$, and fit exactly as for the data (Figure 15). The p-value of an observed statistic is

$$\hat{p} = \frac{r+1}{N+1}, \tag{12.1}$$

where r of the N simulations are at least as extreme (North et al., 2002); $N = 3\,000$ for SMICA and 1 000 for NILC. We also form a Bayes factor B with uniform priors on the amplitudes

($|\Delta T| \leq 545\,\mu$K) and on the configurations, evaluated in closed form with the Gaussian integral and calibrated in the same way.

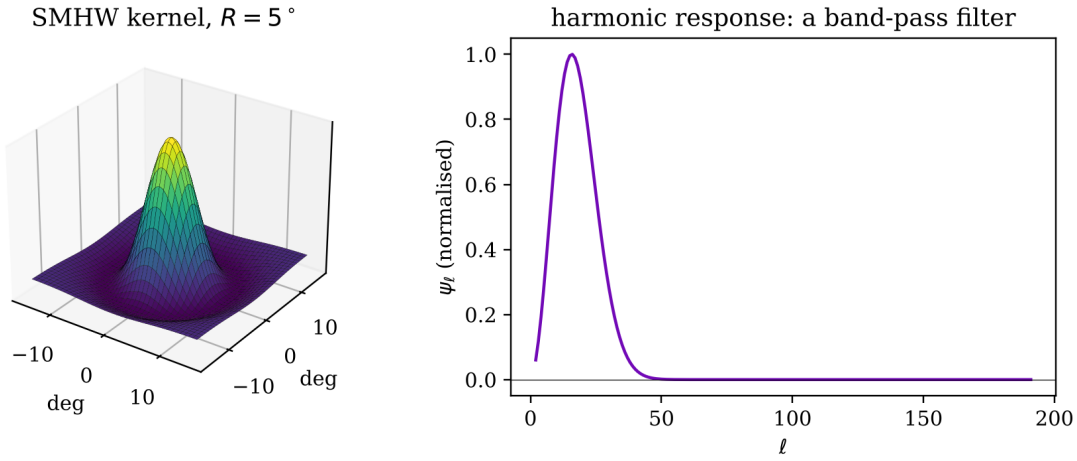


Figure 14. The spherical Mexican-hat wavelet at $R = 5°$ used to select the coldest spot. Left: the kernel as a surface, positive core and negative ring. Right: its normalised Legendre coefficients, a band-pass filter peaking near $\ell \approx 16$.

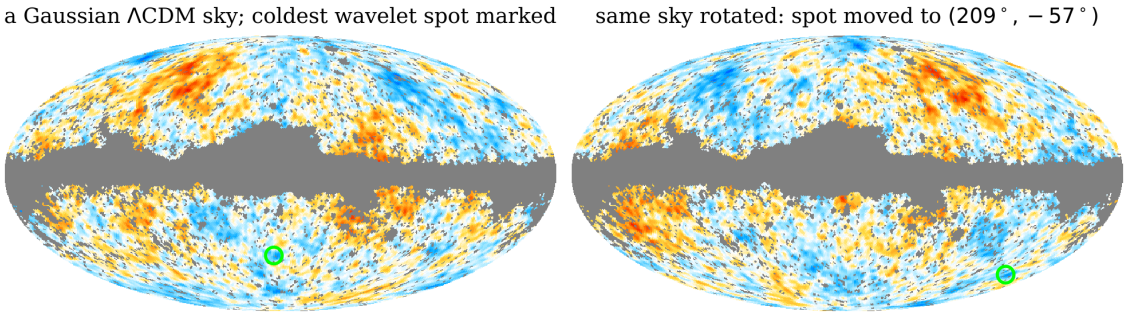


Figure 15. The selected-position null on one simulated sky. Left: a Gaussian ΛCDM realisation with the Planck mask; the circle marks its coldest wavelet spot. Right: the same sky rotated so that this spot moves to the Cold Spot position.

12.3. Results

The minimum of the $R = 5°$ wavelet coefficient over the unmasked SMICA sky is $-12.4\,\mu$K at $(202.5°, -55.9°)$, 4° from the nominal centre. Gaussian skies have a minimum at least this low with probability 0.36: at this scale the Cold Spot is the coldest feature of the sky but not by itself an outlier.

Table 1 lists the best fits and Figure 16 shows them. The best disc has $\theta_c = 16°$, centre amplitude $z_0 = -3.35 \times 10^{-5}$ ($-91\,\mu$K) and edge amplitude $z_{\rm crit} = +1.63 \times 10^{-5}$ ($+44\,\mu$K), and improves the fit by $\Delta\chi^2 = 23.1$. The best texture has $\theta_c = 6°$ and $\varepsilon = -7.92 \times 10^{-5}$, with $\Delta\chi^2 = 19.0$, close to the WMAP value of Cruz et al. (2007). NILC gives the same within the Monte Carlo error.

Table 2 shows how the significance falls as more of the freedom actually used is included, and Figure 17 shows the null distributions. For one template chosen in advance the improvements would correspond to more than four standard deviations. At a fixed position, with the maximum over sizes and centres, they correspond to about three. Once the selection of the coldest spot is reproduced they fall to $p = 0.050$ (disc) and $p = 0.030$ (texture), about two standard deviations. A Bayes factor computed at a spot chosen for its extremeness inherits the selection: $\ln B = 3.74$ for the texture would count as strong evidence on the usual scales, but selected Gaussian skies reach it with probability 0.015.

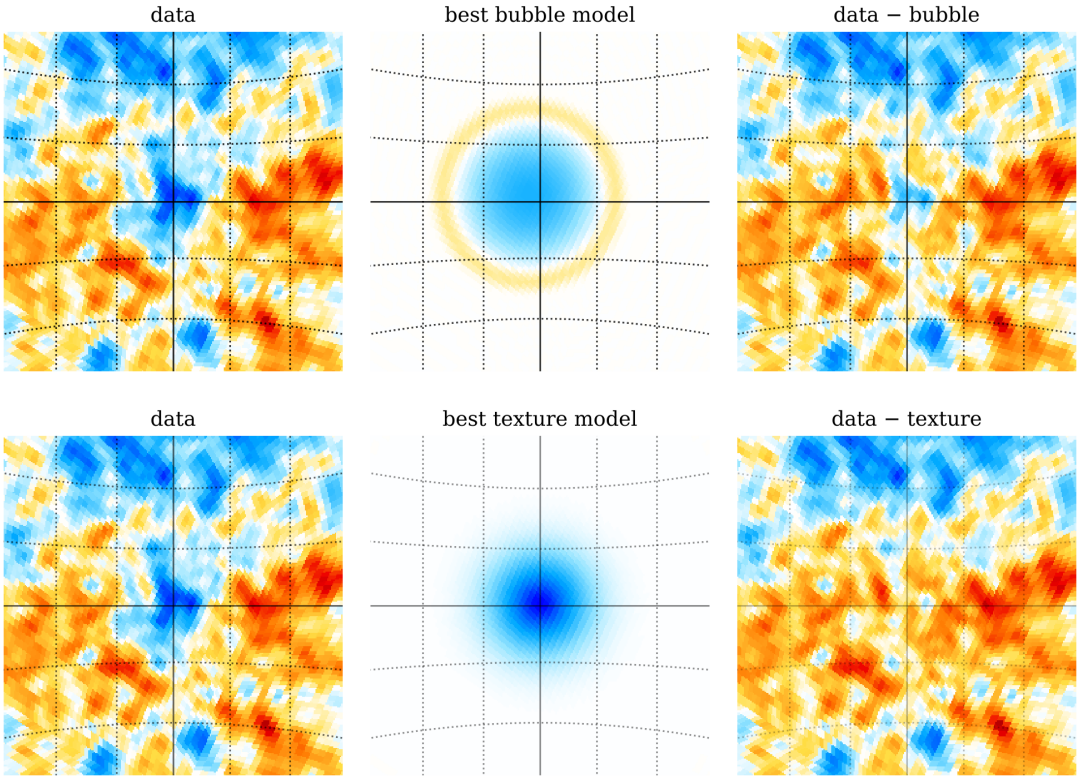


Figure 16. Data, best model and residual, $61° \times 61°$ around the Cold Spot, common colour scale $\pm 200\,\mu$K. Upper row: disc, $\theta_c = 16°$; the warm edge of the disc is the bright ring of the model. Lower row: texture, $\theta_c = 6°$. Both remove the cold core; neither removes the surrounding structure.

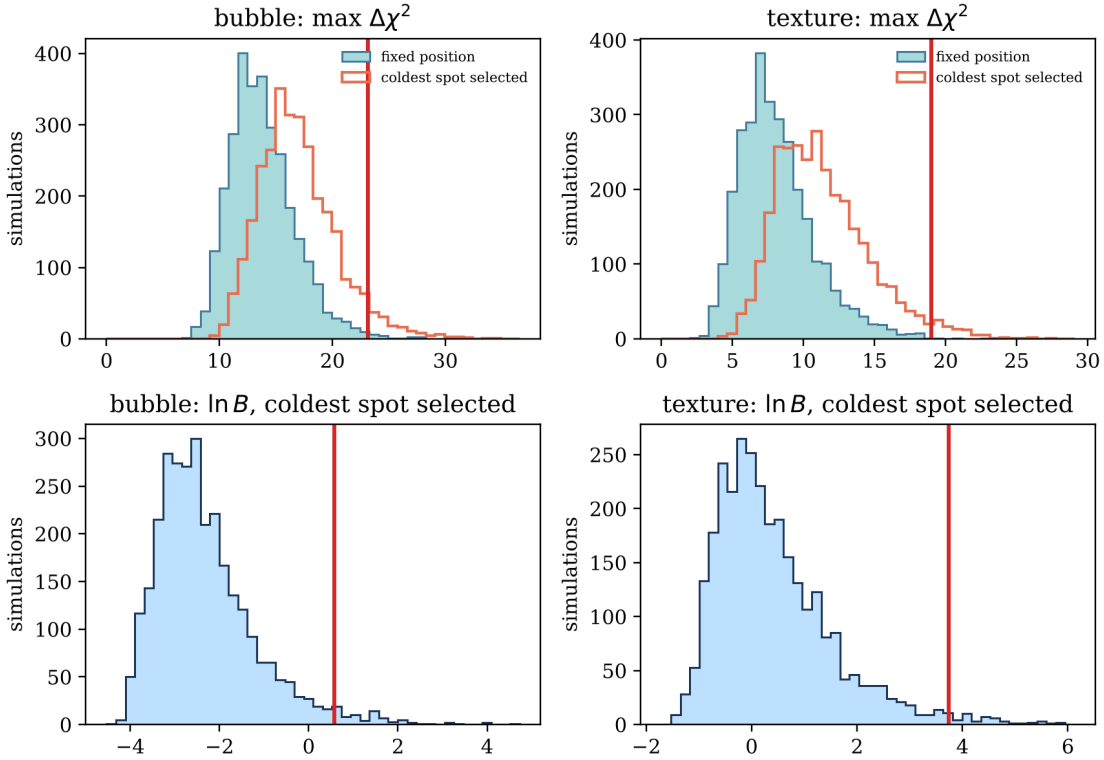


Figure 17. Null distributions from 3 000 Gaussian skies. Upper row: largest $\Delta\chi^2$ at a fixed position and at the selected coldest spot. Lower row: $\ln B$ at the selected spot. Vertical lines: Planck SMICA.

Table 1. Best-fitting templates at the Cold Spot. $p_{\rm fix}$ and $p_{\rm sel}$: probability that a Gaussian sky gives a larger maximum $\Delta\chi^2$ at a fixed position, or at its selected coldest spot. p_B: the same for the Bayes factor at the selected spot.

Map	Model	θ_c	amplitudes	$\Delta\chi^2$	$p_{\rm fix}$	$p_{\rm sel}$	$\ln B$ (p_B)
SMICA	disc (B)	16°	$z_0 = -3.35 \times 10^{-5}$, $z_{\rm crit} = 1.63 \times 10^{-5}$	23.1	0.0073	**0.050**	0.57 (0.026)
SMICA	texture	6°	$\varepsilon = -7.92 \times 10^{-5}$	19.0	0.0017	**0.030**	3.74 (0.015)
NILC	disc (B)	16°	$z_0 = -3.34 \times 10^{-5}$, $z_{\rm crit} = 1.64 \times 10^{-5}$	23.4	0.0100	**0.065**	0.56 (0.032)
NILC	texture	6°	$\varepsilon = -7.90 \times 10^{-5}$	18.9	0.0030	**0.023**	3.71 (0.017)

Table 2. From nominal to calibrated significance at the Cold Spot (SMICA). Gaussian equivalents are two-sided.

	disc		texture	
reference distribution	p	σ	p	σ
one template fixed in advance (χ^2_k)	9.6×10^{-6}	4.4	1.3×10^{-5}	4.4
max over sizes and centres, fixed position	0.0073	2.7	0.0017	3.1
max over sizes and centres, coldest spot selected	0.050	2.0	0.030	2.2

Three freedoms are not in these numbers and would each raise the p-values: the Cold Spot was originally picked among several wavelet scales and among hot and cold extremes; the template families were chosen after the Cold Spot was known; and the disc preference is a sharp peak at 16° (Figure 18), with $\Delta\chi^2 = 14.7$ at 14° and 1.2 at 18°. The preference rests on the position of the warm ring around the Cold Spot (Figure 19), and the fitted edge jump $z_{\rm crit} \neq 0$ means that the data favour the two-amplitude disc over the simplest profile of Proposition 10.1. The targeted test therefore gives a preference of about two standard deviations, and no detection.

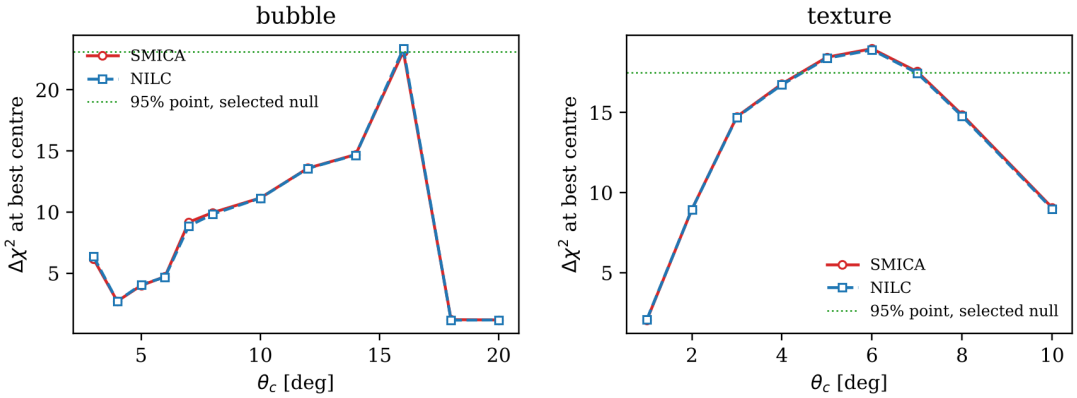


Figure 18. Improvement $\Delta\chi^2$ at the best centre as a function of template size, SMICA (circles) and NILC (squares). The dotted line is the 95% point of the selected-position null for the maximum over all sizes.

13. The blind search of the whole sky

13.1. Why a blind search

A test at the Cold Spot cannot tell whether the sky holds imprints elsewhere, and the scenario f_2 of (7.1) predicts several. The second test scans every position and size on the unmasked sky of all four maps and calibrates the largest value found with simulations that repeat the whole scan. Figure 20 summarises it.

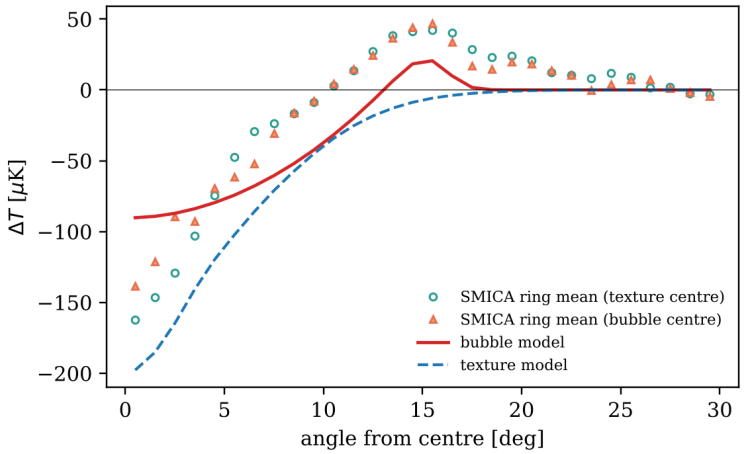


Figure 19. Mean SMICA temperature in 1° rings around the best texture centre (circles) and the best disc centre (triangles), with the smoothed best-fit models. The cold core and the warm ring at 12–16° are what both templates describe.

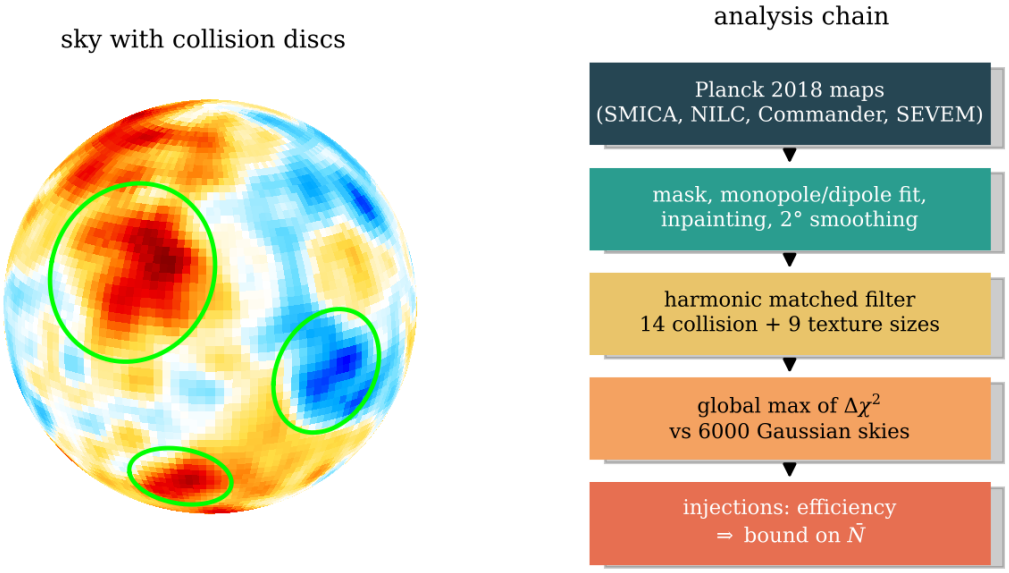


Figure 20. Left: a sketch of a sky with three disc imprints (rings); colours are a smooth random field plus disc-shaped offsets of both signs. Right: the chain of the blind search.

13.2. A matched filter that is exact on the sphere

For the full sky the mask is kept where the degraded value exceeds 0.9 ($f_{\rm sky} = 0.75$), and masked pixels are filled by diffusion before the harmonic transform.[4] A zonal profile $f(\theta)$ has Legendre coefficients $f_\ell = 2\pi \int_{-1}^{1} f\, P_\ell \, d\cos\theta$. The GLS quantities of Proposition 11.1, for an isotropic sky, become

(13.1)

$$b_k(\hat{\mathbf{n}}_0) = \sum_{\ell m} \frac{f_{k\ell} T_\ell}{S_\ell}\, d_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}_0), \qquad A_{kk'} = \sum_{\ell \geq 2} \frac{2\ell+1}{4\pi}\, \frac{f_{k\ell} f_{k'\ell} T_\ell^2}{S_\ell}, \qquad \Delta\chi^2(\hat{\mathbf{n}}_0) = \boldsymbol{b}^{\top} \mathsf{A}^{-1} \boldsymbol{b}$$

(Appendix B). The vector $\boldsymbol{b}$ is a filtered map, so one harmonic synthesis per template component gives $\Delta\chi^2$ at every pixel, and A does not depend on position. We scan 14 disc radii, $\theta_c = 3, 4, 5, 6, 8, 10, 12, 14, 16, 18, 20, 23, 26, 30$ degrees, and 9 texture scales, $1, \ldots, 8, 10$ degrees. The

[4] Each masked pixel takes the mean of its eight neighbours, repeated 400 times. Zero-filling instead produces spurious $\Delta\chi^2 \approx 100$ at the edges of point-source holes. No masked pixel enters a detection (validity rule below).

resulting one-sigma errors on the central temperature are 30–42 μK for the SW disc and 49–77 μK for the texture. As a check, the texture error at 6° from (13.1), $\sigma_\varepsilon = 1.82 \times 10^{-5}$, equals the pixel-space value of the targeted test, and the closed-form coefficients agree with numerical quadrature to 3×10^{-11} (discs) and 10^{-9} (textures).

13.3. Valid centres

Near the mask the filter output mixes in inpainted pixels. A centre is accepted for size θ_c only if at least 85 per cent of the pixels within $\theta_c + 2°$ are unmasked. The accepted fraction of the sky falls from 68 per cent for the smallest sizes to 43 per cent for $\theta_c = 30°$ (Figure 21). The rule is applied identically to data, simulations and injections.

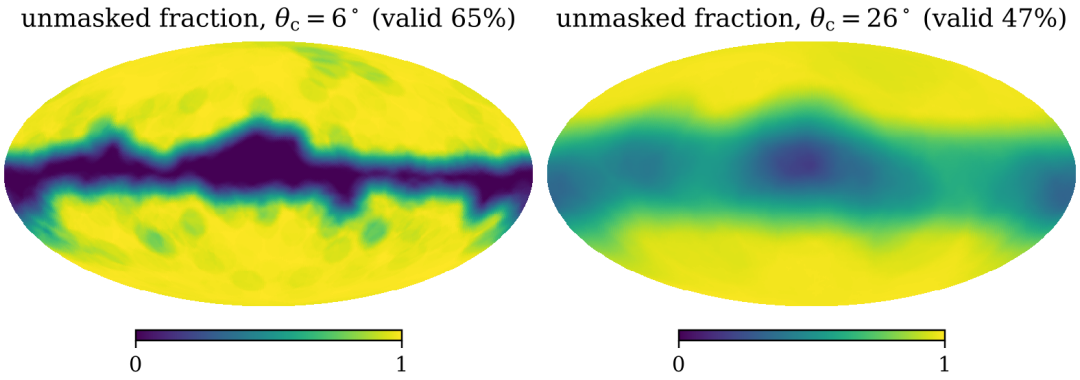


Figure 21. Unmasked fraction inside a disc of radius $\theta_c + 2°$ around each centre, for $\theta_c = 6°$ and 26°. Centres below 0.85 are excluded.

13.4. Global statistic

For each family we take $D(\hat{\mathbf{n}}_0) = \max_{\theta_c} \Delta\chi^2(\hat{\mathbf{n}}_0; \theta_c)$ and its maximum $D_{\max}$ over all valid centres, for the SW disc ($D_{\rm SW}$), the two-amplitude disc ($D_{\rm B}$) and the texture ($D_{\rm T}$). Each is calibrated with 6 000 Gaussian skies passed through the same mask, monopole and dipole fit, inpainting, filter and validity rule; p follows (12.1). The threshold τ_{95} is the 95th percentile of the simulated $D_{\max}$: a Gaussian sky produces anything above it anywhere with probability 5 per cent. We find $\tau_{95} = 25.2$ (SW), 34.8 (B) and 24.6 (T). A candidate is a peak of D above τ_{95}; peaks closer than 8° are merged.

13.5. Results

Figure 22 shows $D_{\rm SW}$, $D_{\rm B}$ and $D_{\rm T}$ for SMICA. The maps look like Gaussian skies: many moderate peaks of $\Delta\chi^2 \approx 10$–20, no isolated outlier, and best-fitting sizes that change from one structure to the next (Figure 23). Table 3 gives $D_{\max}$, the global p-value and the number of candidates for every map and family. All twelve p-values lie between 0.31 and 0.67, and there are no candidates (Figure 24). The four maps agree to within 0.3 in $D_{\max}$ and pixel by pixel (Figure 25), so differences in foreground cleaning do not drive the result.

Table 3. Global statistics of the blind search: largest $\Delta\chi^2$ over valid centres and sizes, global p-value from 6 000 simulations and number of separated peaks above τ_{95}.

	SW disc			two-amplitude disc			texture		
Map	$D_{\max}$	p	n_{95}	$D_{\max}$	p	n_{95}	$D_{\max}$	p	n_{95}
SMICA	21.0	0.32	0	26.0	0.63	0	18.9	0.55	0
NILC	20.8	0.35	0	25.7	0.67	0	18.8	0.57	0
Commander	20.9	0.34	0	25.8	0.67	0	19.0	0.53	0
SEVEM	21.1	0.31	0	25.8	0.66	0	19.1	0.52	0
τ_{95}	25.2			34.8			24.6		
τ_{99}	29.2			40.9			28.2		

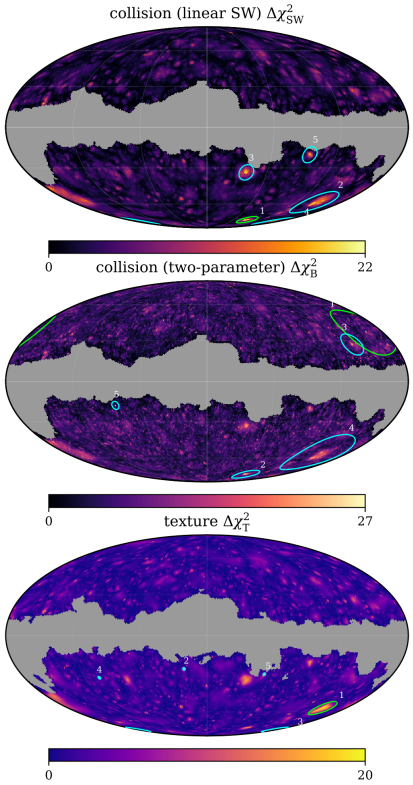


Figure 22. Maximum over sizes of $\Delta\chi^2$ for SMICA: SW disc (top), two-amplitude disc (middle) and texture (bottom). Excluded centres are grey. Circles mark the five strongest separated peaks at their best radius; the strongest is green. The Cold Spot is the strongest texture peak (bottom, number 1).

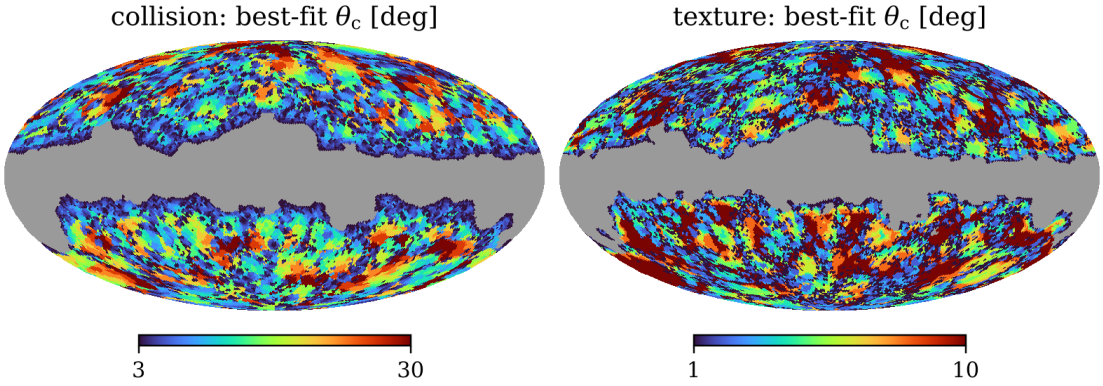


Figure 23. Best-fitting radius at each valid centre for the SW disc (left) and the texture (right) in SMICA. The winning size changes from one structure to the next, as in a Gaussian sky.

Table 4 lists the three strongest peaks of each family in SMICA, and Figure 26 shows six of them. Two features recur. One is a cold disc of radius 4°–5° near the south Galactic pole at $(268°, -77°)$. The other is the Cold Spot: the strongest texture peak, the second SW peak (as a 10° cold disc centred 3° away) and the fourth two-amplitude peak.[5] Its texture fit gives $\varepsilon = (-7.9 \pm 1.8) \times 10^{-5}$, a central temperature of $-215 \pm 50\,\mu$K. With the search over the sky and sizes counted, its $D_{\rm T} = 18.9$ is just below the simulated median of 19.2: $p = 0.55$.

[5]In SEVEM and Commander the Cold Spot is the strongest SW peak; the two leading SW peaks differ by less than 0.4 in every map.

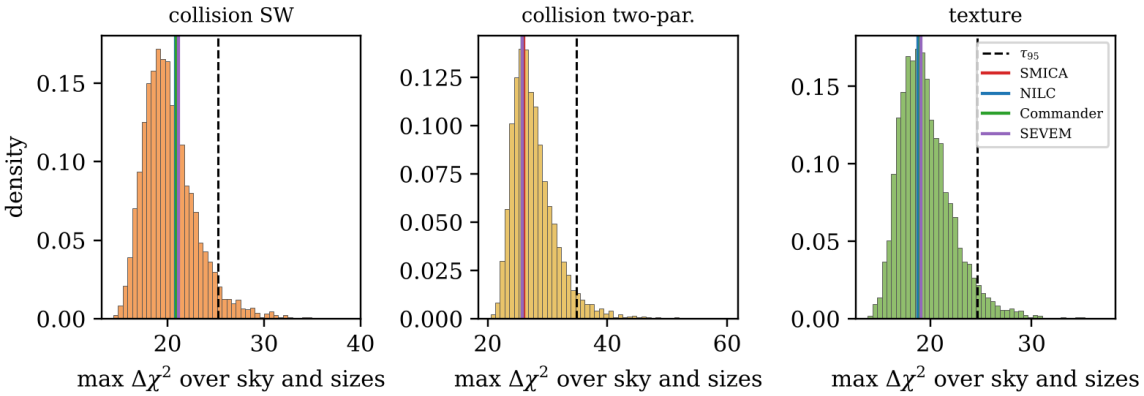


Figure 24. Distributions of the global maximum $D_{\max}$ in 6 000 Gaussian skies (histograms), with τ_{95} (dashed) and the observed values of the four maps (coloured lines).

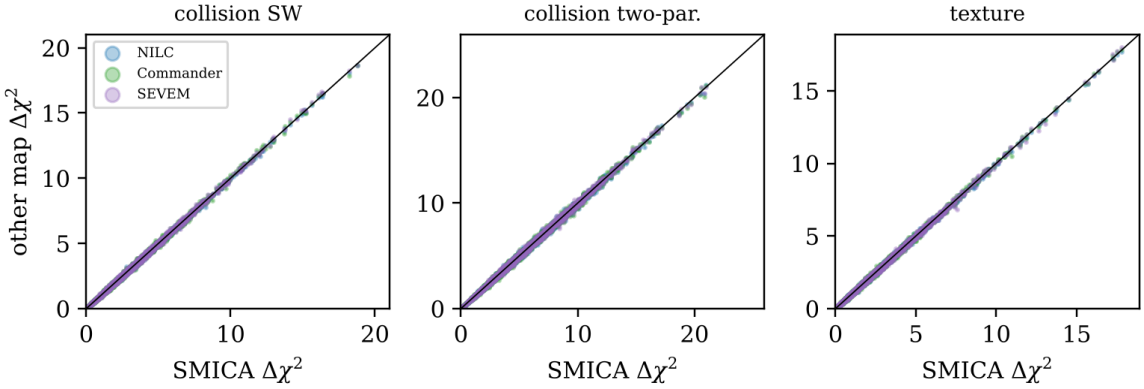


Figure 25. Pixel-by-pixel comparison of the $\Delta\chi^2$ maps of NILC, Commander and SEVEM against SMICA (every seventh valid pixel).

Table 4. Strongest separated peaks in SMICA: position, best radius, $\Delta\chi^2$ and fitted central amplitude (for the two-amplitude disc: central and edge amplitudes).

l [deg]	b [deg]	θ_c [deg]	$\Delta\chi^2$	amplitude [μK]
SW disc				
268	−77	4	21.0	-175 ± 38
213	−59	10	20.7	-144 ± 32
321	−34	6	19.7	$+152 \pm 34$
two-amplitude disc				
196	+37	18	26.0	$+4 \pm 31$, -55 ± 11
267	−78	5	25.7	-150 ± 36, $+78 \pm 24$
220	+27	8	24.2	$+130 \pm 33$, -53 ± 18
texture				
209	−57	6	18.9	-215 ± 50
22	−25	1	17.3	-321 ± 77
183	−79	3	16.5	-220 ± 54

13.6. The Galactic plane

Removing in addition all pixels with $|b| < 20°$ and recalibrating with 1 500 simulations processed with the same cut gives global p-values between 0.27 and 0.67 and no candidates (Figure 27). The result does not depend on the regions near the plane.

14. Detection efficiency and the bound on the number of imprints

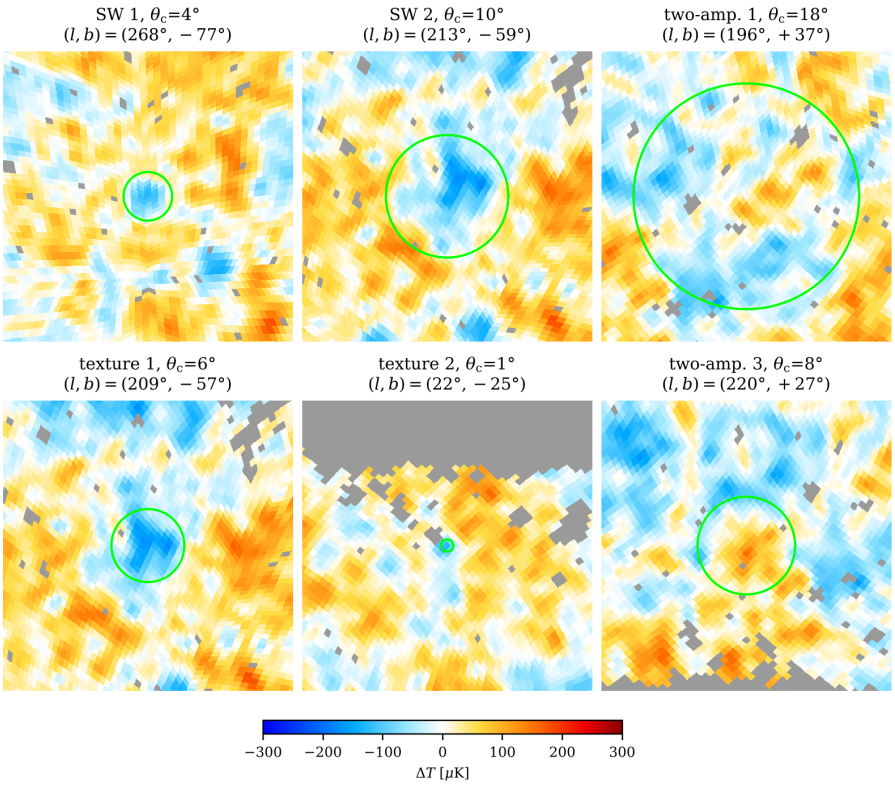


Figure 26. The SMICA map around six of the strongest peaks of Table 4, gnomonic projection 48° across, with the best radius in green. Panels SW 2 and texture 1 are both the Cold Spot. None is significant after the search is counted.

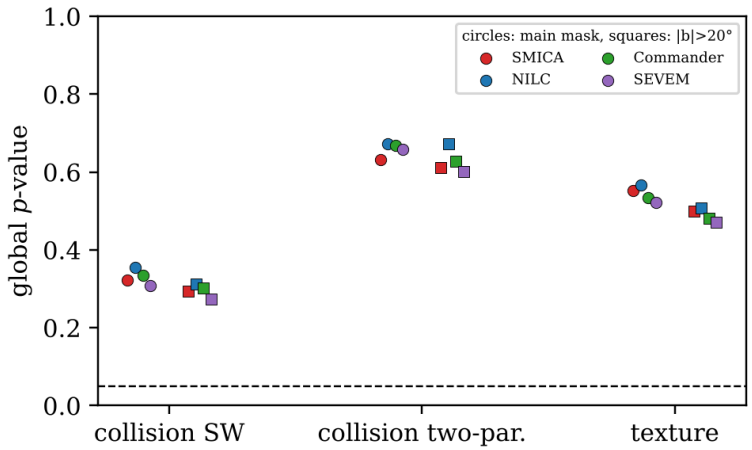


Figure 27. Global p-values for the three families and four maps, with the main mask (circles) and with an additional $|b| > 20°$ cut (squares). The dashed line marks $p = 0.05$.

14.1. Injections

To turn the absence of candidates into a bound we add one synthetic imprint to each of many simulated skies, with centre uniform on the sphere, random sign and size drawn from a prior ($\propto \sin\theta_c$ on the disc grid, uniform on the texture grid). Discs are injected with the SW profile (10.2) at central temperature $|\Delta T_0| = T_0|z_0|$, textures at $T_0|\varepsilon|$. An injection is recovered if the statistic exceeds τ_{95} within $\max(3°, \theta_c/2)$ of the true centre. The recovered fraction is the efficiency $\eta(|\Delta T_0|)$; it includes the loss to the mask. In total 9 600 imprints were injected, 300 per amplitude, plus a grid of 100 per cell at fixed radius. Figure 28 shows one.

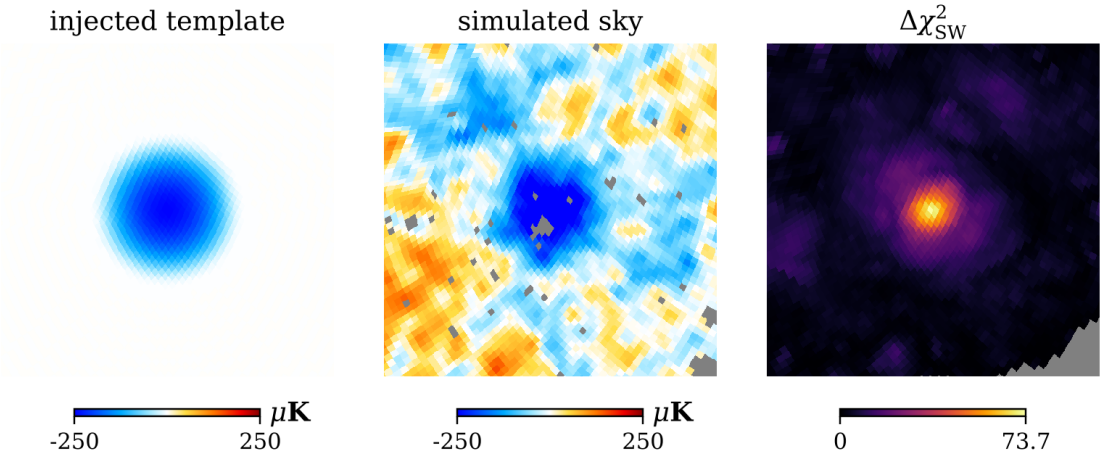


Figure 28. An injected SW disc with $\theta_c = 12°$ and $\Delta T_0 = -250\,\mu$K at $(60°, 45°)$: the template (left), template plus a Gaussian sky with the point-source mask in grey (centre) and the recovered $\Delta\chi^2_{\rm SW}$ (right), which reaches 73.7 at the centre against $\tau_{95} = 25.2$.

14.2. Efficiency

Table 5 and Figure 29 give the efficiencies. For discs η rises from 0 at $60\,\mu$K to 0.75 at $400\,\mu$K; for textures from 0 at $100\,\mu$K to 0.77 at $600\,\mu$K. The plateau is set by the mask: among injections centred at valid positions the efficiency at the highest amplitude is 1.00 for discs and 0.99 for textures, as Proposition 10.3 requires. Figure 30 resolves the disc efficiency by radius: small discs need larger amplitudes because their errors are larger, large discs lose valid sky. At $\theta_c \geq 8°$ an amplitude of $210\,\mu$K is recovered in most valid positions.

Table 5. Detection efficiency η (binomial error) and the 95 per cent upper limit on the expected number of imprints on the full sky, as a function of central temperature.

SW discs			textures		
$\lvert\Delta T_0\rvert$ [μK]	η	$\bar{N}_{95}$	$\lvert\Delta T_0\rvert$ [μK]	η	$\bar{N}_{95}$
60	0.00 ± 0.00	–	100	0.00 ± 0.00	898.7
90	0.05 ± 0.01	64.2	150	0.03 ± 0.01	89.9
120	0.10 ± 0.02	30.0	200	0.15 ± 0.02	20.4
150	0.30 ± 0.03	10.0	250	0.36 ± 0.03	8.4
180	0.47 ± 0.03	6.3	300	0.54 ± 0.03	5.5
210	0.63 ± 0.03	4.7	350	0.66 ± 0.03	4.6
250	0.68 ± 0.03	4.4	400	0.68 ± 0.03	4.4
300	0.68 ± 0.03	4.4	500	0.77 ± 0.02	3.9
400	0.75 ± 0.02	4.0	600	0.77 ± 0.02	3.9

14.3. Bounds

If imprints are independent and $\bar{N}$ is their expected number on the sphere, the number detected is Poisson with mean $\eta\bar{N}$, plus false positives, which we ignore to stay conservative. With no candidates the 95 per cent upper limit on a Poisson mean is $\lambda_{95} = -\ln 0.05 = 3.00$ (Appendix C), so

$$\bar{N} < \frac{3.00}{\eta(|\Delta T_0|)}. \tag{14.1}$$

This gives the limits of Table 5. At $|\Delta T_0| \geq 210\,\mu$K ($|z_0| \geq 7.7 \times 10^{-5}$) the expected number of disc imprints on the full sky is below 4.7; at $|\Delta T_0| \geq 350\,\mu$K ($|\varepsilon| \geq 1.3 \times 10^{-4}$) the expected number of textures is below 4.6. For discs of fixed radius $\theta_c \geq 8°$ and $|\Delta T_0| \geq 300\,\mu$K the bound is $\bar{N} < 4.5$. These values agree with the WMAP bounds of $\bar{N} < 4$ for collisions and 5.2 for textures (Feeney et al., 2013), as expected: on these scales temperature maps are limited by

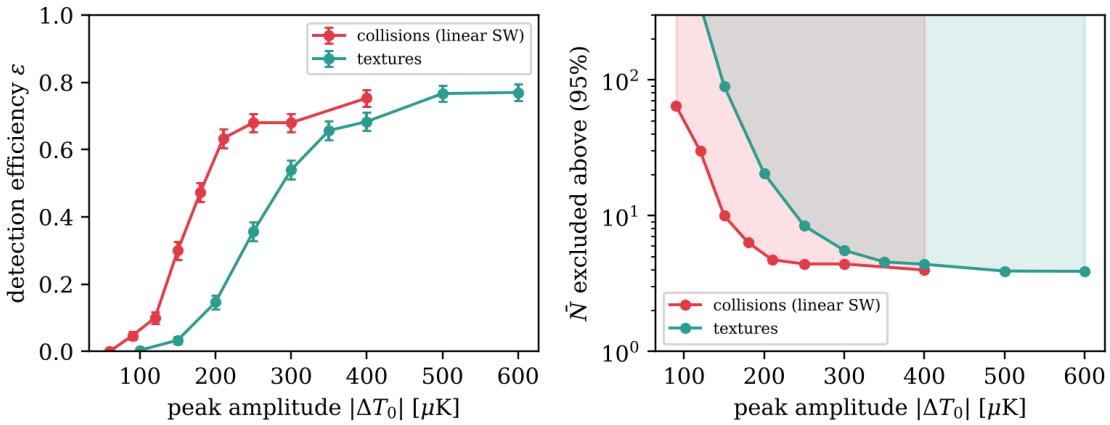


Figure 29. Left: detection efficiency against central temperature for discs (size prior $\propto \sin\theta_c$) and textures (uniform). Right: the 95 per cent upper limit on the expected number of imprints on the full sky; shaded regions are excluded.

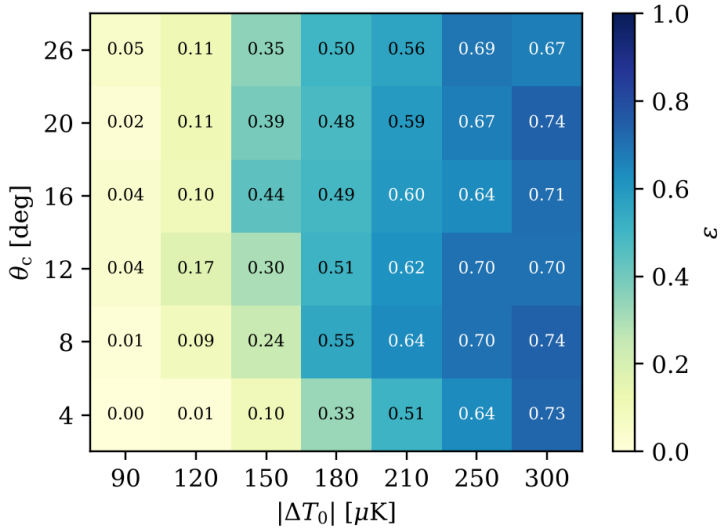


Figure 30. Disc efficiency at fixed radius and amplitude (100 injections per cell).

cosmic variance, which WMAP already reached (Feeney et al., 2015). The Planck result rests on four independent component separations.

For the conjugate-universe framework the bound applies to scenario f_2: periodic encounters that left more than five bright imprints are excluded. Scenario f_1 predicts a single imprint, for which a Poisson bound is not the right tool; Section 15 treats it directly.

15. The observable window

Figure 31 draws the window of Definition 10.2 with the measured τ_{95} and $\sigma_A(\theta_c)$, and Table 6 lists the edges. The detection edge of the SW disc lies at 154–211 μK, lowest for radii above 12°; the closure line at 203–280 μK. For textures the edge lies at 244–383 μK and the closure line at 324–511 μK. The efficiencies of Section 14, which include the mask and the matching of positions, agree: at valid positions the disc efficiency reaches 1.00 and the texture efficiency 0.99 at the highest injected amplitudes.

The figure shows three things.

Bright imprints are excluded. A conjugate-universe imprint with a disc radius between 3° and 30° and a central temperature above about 210 μK, or a texture between 1° and 10° above about 330 μK, would have been seen with at least 95 per cent probability if centred on the 43–68 per cent of the sky that the search accepts. None is seen. For scenario f_1, with exactly one

Table 6. The window edges: detection edge A_{50} and closure line A_{95} (central temperature, μK) for selected radii.

SW disc ($\tau_{95} = 25.2$)			texture ($\tau_{95} = 24.6$)		
θ_c	A_{50}	A_{95}	θ_c	A_{50}	A_{95}
3°	211	280	1°	383	511
4°	191	254	2°	296	394
6°	172	229	4°	256	341
10°	159	211	6°	246	328
16°	154	204	8°	244	324
30°	157	209	10°	245	326

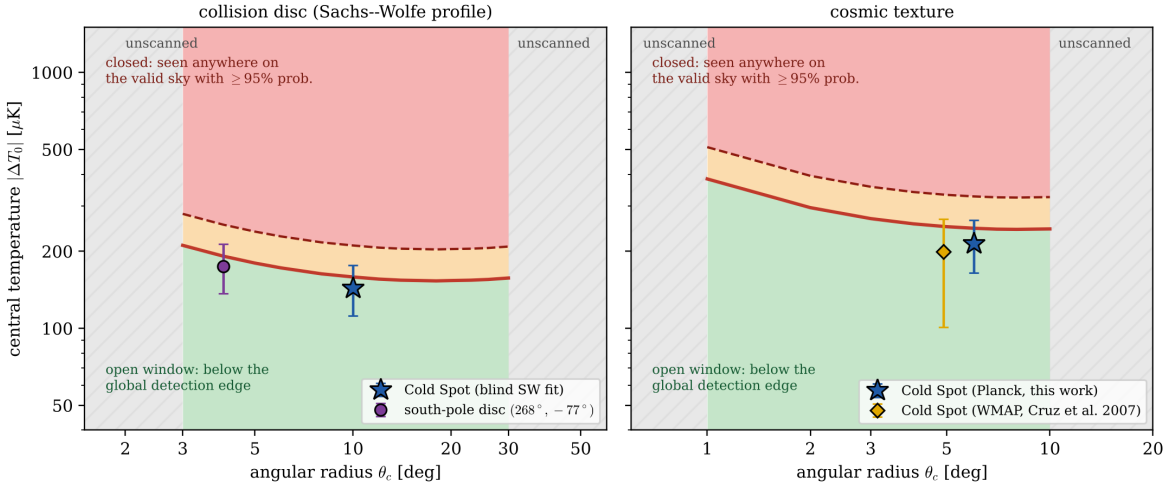


Figure 31. The observable window for a parallel-universe imprint in Planck 2018 temperature data. Red: closed, an imprint would have been seen anywhere on the valid sky with probability at least 95 per cent. Orange: between the detection edge (solid) and the closure line (dashed). Green: open. Hatched: sizes not scanned. Left: SW discs, with the Cold Spot as a 10° disc (star) and the south-pole disc (circle). Right: textures, with the Planck fit to the Cold Spot (star) and the WMAP fit of Cruz et al. (2007) (diamond).

imprint at an unknown position, the probability that such an imprint escaped is at most the fraction of the sky the search does not accept plus the 5 per cent miss probability on accepted sky, about 0.6 in all; at the Cold Spot itself, which lies on accepted sky, it is at most 0.05.

The Cold Spot sits just inside the edge. As a texture, the Cold Spot has $|\Delta T_0| = 215 \pm 50\,\mu$K at $\theta_c = 6°$, against an edge of 246 μK: it lies 13 per cent below the amplitude at which the blind search would have flagged it as a global detection. As an SW disc of radius 10° it has $144 \pm 32\,\mu$K against an edge of 159 μK, 9 per cent below. The south-pole disc lies in the same place relative to its edge. This is where the largest fluctuations of a Gaussian sky are expected to lie, since the median of the global maximum, 19.2 for textures, is below $\tau_{95} = 24.6$ by construction. It is also where a real imprint slightly weaker than the detection limit would lie. Present temperature data cannot distinguish the two.

The window is still open. Imprints fainter than the edge, smaller than 3° or larger than 30°, and imprints centred in the regions excluded by the mask are not constrained. The conjugate universe of Part I, with the junction imprint at the Cold Spot, is consistent with the data at the measured amplitude; it is not required by them.

15.1. Simulating a real imprint at the Cold Spot amplitude

The edge of Proposition 10.3 is a lower bound. To see how often the blind search would actually catch a real imprint as bright as the Cold Spot, we simulate it directly. For each of

nine amplitudes between 100 and 500 μK we draw 1000 Gaussian skies from the null model of Section 11, add a 6° texture with central temperature $-|\Delta T_0|$ at a centre drawn uniformly on the sphere, and run the complete texture search: masking, all nine sizes, the validity rule and the fixed threshold $\tau_{95} = 24.6$ calibrated on the null simulations.[6] An imprint counts as detected if $D_{\mathrm{T}} > \tau_{95}$ anywhere within 3° of its true centre, the matching radius of Section 14. Table 7 and Figure 32 give the result.

Table 7. Blind detection of a simulated 6° texture imprint, 1000 skies per amplitude. P_{all}: any centre; P_{valid}: centres on accepted sky (binomial error); bound: Proposition 10.3; $\tilde{D}$: median of max D_{T} within 3° of the centre, valid centres.

$\lvert\Delta T_0\rvert$ (μK)	P_{all}	P_{valid}	bound	$\tilde{D}$
100	0.004	0.006 ± 0.003	0.00	8.3
150	0.042	0.064 ± 0.010	0.03	13.0
180	0.101	0.150 ± 0.014	0.09	16.0
215	**0.266**	$\mathbf{0.398 \pm 0.019}$	**0.27**	**22.5**
246	0.422	0.600 ± 0.019	0.50	26.9
280	0.605	0.844 ± 0.014	0.75	33.7
320	0.686	0.947 ± 0.009	0.93	43.4
400	0.731	0.998 ± 0.002	1.00	67.2
500	0.747	1.000 ± 0.000	1.00	102.9

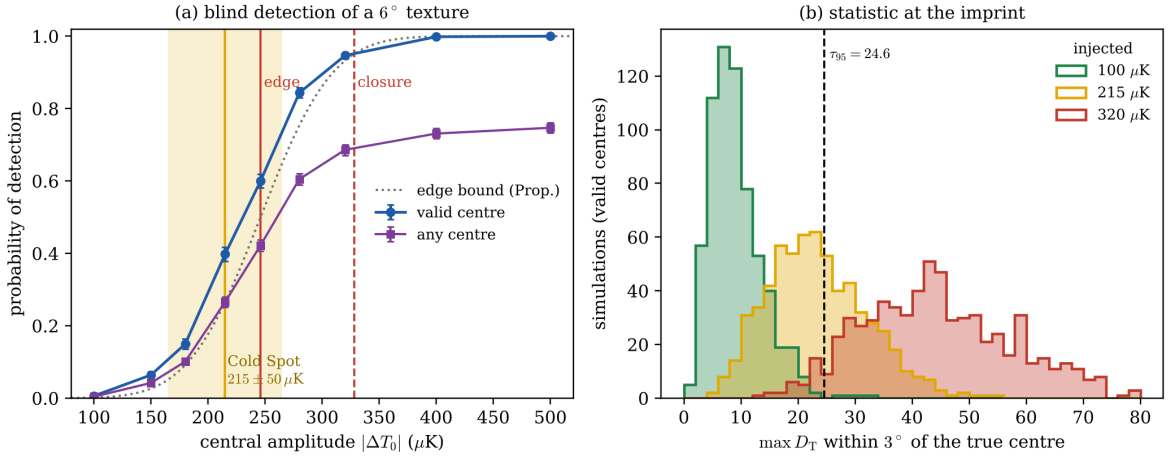


Figure 32. Simulated Cold Spot imprints. (a) Probability that the blind search detects a 6° texture of central amplitude $|\Delta T_0|$, for centres on accepted sky (blue) and anywhere (purple), with the lower bound of Proposition 10.3 (dotted). The gold band is the Planck fit to the Cold Spot, $215 \pm 50\,\mu$K. (b) Distribution of the texture statistic at the imprint for three amplitudes; the dashed line is the global threshold.

Four facts follow.

A real Cold Spot imprint would usually be missed. At 215 μK the search detects the imprint in 39.8 ± 1.9 per cent of the skies in which its centre lies on accepted sky, and in 26.6 per cent of all skies. A conjugate-universe imprint of exactly the fitted strength would therefore escape the blind search about three times in five even where the sky is clean. The blind non-detection of the Cold Spot is what such an imprint would produce in most universes; it lowers the odds of the imprint hypothesis by a factor of only 0.60.[7]

[6] The threshold is not re-tuned; it is the one used on the data. Each amplitude has its own random seed (501–509), and the code is in the repository listed under Data availability.

[7] Under the null hypothesis the Cold Spot fails the global threshold with probability close to one; under the imprint hypothesis with probability 0.60. The likelihood ratio of the observed non-detection is therefore about 0.6, far from decisive.

The analytic edge is confirmed and is conservative. The measured probabilities lie above the bound $\Phi(|A|/\sigma_A - \sqrt{\tau_{95}})$ at every amplitude, as Proposition 10.3 requires, by 0.06–0.13 near the edge: scanning nine sizes and 3° of positions gives the imprint more chances to cross the threshold than a single fixed filter. At 246 μK, the edge, the probability is 0.60 rather than 0.50; at 328 μK, the closure line, it exceeds 0.95.

The mask, not the noise, limits the bright end. Above 400 μK every imprint centred on accepted sky is found, while the all-sky probability levels off at 0.75, close to the accepted fraction of the sky plus the strip within 3° of its border. A bright imprint can hide only behind the Galactic mask or near a point source.

The threshold behaves as calibrated. Exceedances more than 8° from the injected centre occur in 4.9 per cent of the skies at 100 μK and in 4.4–6.0 per cent up to 246 μK, as the 95 per cent threshold demands. At higher amplitudes the rate grows with the amplitude (0.38 at 500 μK): these are secondary responses of the larger filters to the imprint itself, not independent false alarms.[8]

Together these simulations place the Cold Spot where the window drawing put it: inside the open region, at the amplitude where a real imprint and a Gaussian fluctuation look alike to the blind search. A measurement of the Cold Spot that halves the error on $|\Delta T_0|$, or a second, independent observable such as the polarisation signature discussed below, is needed to move it to either side of the edge.

16. Discussion

16.1. What would close or open the window

Temperature maps on these scales are at the cosmic-variance limit, so a better temperature map will not move the edge (Feeney et al., 2015). Three kinds of data can.

Polarisation. A disc imprint also polarises the CMB, with a characteristic E-mode pattern inside and just outside the disc, correlated with the temperature template in a known way (Czech et al., 2010). A joint temperature–polarisation fit at the Cold Spot lowers the false-positive rate for the same imprint. Large-scale polarisation maps from LiteBIRD (LiteBIRD Collaboration, 2023) and the Simons Observatory (Simons Observatory Collaboration, 2019) make this test possible. The prediction of the framework is specific: the polarisation pattern must match the 16° disc or the 6° texture fitted here, with the same sign.

The foreground void. Three-dimensional surveys of the Eridanus region narrow the integrated Sachs–Wolfe contribution of the supervoid (Szapudi et al., 2015; Kovács et al., 2022). Every microkelvin attributed to the void lowers the amplitude left for a primordial imprint and moves the Cold Spot deeper into the open window, away from detection.

Other imprints. Scenario f_2 predicts further imprints. Any second disc or texture above the edge would be a candidate; the bound $\bar{N} < 4.7$ already limits their number.

16.2. Comparison with other multiverse proposals

In the many-worlds interpretation (Everett, 1957) branches are not spatially separated and leave no imprint on the sky. In the ekpyrotic model (Khoury et al., 2001) the Big Bang is a collision of branes in a higher-dimensional bulk; its imprint is not localised. Bubble collisions in eternal inflation (Aguirre & Johnson, 2011; Kleban, 2011; Wainwright et al., 2014) produce the disc profile of Proposition 10.1. The conjugate-universe framework of Part I differs in its mechanism, a hidden time direction and a topological split, but Section 10 shows that its observable imprint has the same shape. The data therefore test the two on the same footing, and the window of Figure 31 applies to both. The statistical limits on what a single observed sky can establish about unobservable regions (Nandi et al., 2025), and the higher-dimensional settings in which these ideas are usually formulated (Bhattacharjee, 2022b,a; Samal et al., 2026), enter the test only through the amplitudes and sizes measured here.

[8] A bright 6° texture correlates with the 8° and 10° templates centred several degrees away, and the masked harmonic filter spreads part of its response beyond the template radius. The growth with amplitude, absent at low amplitude, shows the origin.

16.3. Limitations

(i) The two-dimensional time, the entangled pair and the topological split of Part I are hypotheses; they are not derived from a theory of quantum gravity. The data test only their imprint.
(ii) The amplitude and size of the imprint are free. The framework could be excluded only for imprints above the closure line, or confirmed only by a detection.
(iii) The simulations use a fixed ΛCDM spectrum and isotropic noise. Residual foregrounds are not simulated; the agreement of four maps and the stability under the Galactic cut show that they do not create or hide a candidate at the level of τ_{95}.
(iv) The efficiency of the two-amplitude disc statistic was not computed; the bounds refer to the SW disc and the texture.

17. Conclusion

Part I restated the conjugate-universe framework: two universes with oppositely oriented loops of imaginary time, whose winding numbers $\pm 2\pi i$ combine to the affine parameter π; an entangled pair born in one Ricci-flow surgery; a symmetry-breaking transition that hides the imaginary direction. It corrected the residues, the texture profile, the field content needed for textures and the identification of the Cold Spot. Its main new result is that the only observable trace of the junction is a disc of the bubble-collision type or, if the junction carries a texture, a texture spot.

Part II measured with the Planck 2018 maps where such a trace can still hide. At the Cold Spot a collision disc of radius 16° is preferred at about two standard deviations once the selection of the spot is reproduced. Over the whole sky no disc and no texture is significant ($p = 0.31$–0.67 in four maps), and more than 4.7 disc imprints brighter than 210 μK and more than 4.6 textures brighter than 350 μK are excluded at 95 per cent confidence. The Cold Spot, interpreted as the junction imprint, lies 13 per cent below the detection edge, inside the window that remains open.

The Planck data do not detect a parallel universe and do not exclude one. Simulated imprints of the fitted strength escape the blind search three times in five, so the absence of a global detection says little about the Cold Spot either way. They fix the window in which the imprint of a conjugate universe must lie, and they show that the Cold Spot sits at its edge, where polarisation data can test it.

Author contributions

D.B. conceived the framework, designed and carried out the Planck analysis and wrote the manuscript. All authors contributed to the theoretical framework of the earlier versions, discussed the results, checked the manuscript and approved the final version.

Data availability

The Planck 2018 maps, mask and best-fit spectrum are public at the Planck Legacy Archive (`https://pla.esac.esa.int`). The analysis code, random seeds and the numerical results quoted here are in the public repository `https://github.com/creelie/wormhole` (directory `parallel-universe-v3`), archived at Zenodo under DOI 10.5281/zenodo.24190776.

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Conflict of interest

The authors declare no conflict of interest.

Appendix A. Derivations for Part I

A.1 Entropy of the singlet

For (4.2), $|\Psi\rangle\langle\Psi| = \frac{1}{2}(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|)$. Tracing over β removes the cross terms, because $\langle 1|0\rangle_\beta = 0$, and leaves $\varrho_\alpha = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \mathbb{1}/2$. Its eigenvalues are $\frac{1}{2}, \frac{1}{2}$, so $S = -2 \cdot \frac{1}{2}\log_2 \frac{1}{2} = 1$. For $|\Psi(\vartheta)\rangle = \cos\vartheta|01\rangle - \sin\vartheta|10\rangle$ the same computation gives $\varrho_\alpha = \mathrm{diag}(\cos^2\vartheta, \sin^2\vartheta)$ and (4.4).

A.2 Distance between geodesics on the sphere

On the unit sphere let two great circles leave $(1, 0, 0)$ with angle τ between them. At arc length t they are at $p_0 = (\cos t, \sin t, 0)$ and $p_1 = (\cos t, \sin t \cos\tau, \sin t \sin\tau)$, so $\cos d = p_0 \cdot p_1 = \cos^2 t + \sin^2 t \cos\tau$. Writing $\cos\tau = 1 - 2s^2$ with $s = \sin(\tau/2)$, $\cos d = 1 - 2s^2 \sin^2 t$ and

$$\sin^2 d = 1 - \cos^2 d = 4s^2 \sin^2 t\,(1 - s^2 \sin^2 t).$$

On the other hand $\sin^2 t \sin^2\tau\,(1 + \cos^2 t \tan^2\frac{\tau}{2}) = 4s^2 \sin^2 t\,[(1 - s^2) + s^2\cos^2 t] = 4s^2 \sin^2 t\,(1 - s^2 \sin^2 t)$. The two agree, which proves (6.4) whenever $d \le \pi/2$.

A.3 The shrinking neck

Near a neck pinch the metric is close to $g = ds^2 + r(t)^2\, d\Omega_2^2$ on $\mathbb{R} \times S^2$. The Ricci tensor of this product has no component along ds^2, and on the sphere factor equals the round metric $d\Omega_2^2$ (the Ricci tensor of a round two-sphere of radius r is $g_{S^2}/r^2 = d\Omega_2^2$). The flow (5.1) therefore gives $\partial_t(r^2) = -2$, so $r^2(t) = 2(T - t)$ and the neck vanishes at $t = T$. Parabolic rescaling (5.2) keeps the radius of order one and gives the round cylinder as the limit.

Appendix B. The matched filter

Lemma B.1: Addition theorem

$\sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{\mathbf{n}}) Y_{\ell m}^*(\hat{\mathbf{n}}') = \frac{2\ell+1}{4\pi} P_\ell(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}')$.

Proof. The left side is the kernel of the orthogonal projection onto degree-ℓ harmonics and does not depend on the basis; rotations preserve that space, so the kernel depends only on $\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}'$. For fixed $\hat{\mathbf{n}}'$ it is a degree-ℓ harmonic invariant under rotations about $\hat{\mathbf{n}}'$, hence $c_\ell P_\ell(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}')$. Setting $\hat{\mathbf{n}} = \hat{\mathbf{n}}'$ and integrating over the sphere gives $2\ell + 1 = 4\pi c_\ell$. □

Covariance. By the lemma, $\langle \Delta T(\hat{\mathbf{n}}_i) \Delta T(\hat{\mathbf{n}}_j)\rangle = \sum_\ell C_\ell T_\ell^2 \frac{2\ell+1}{4\pi} P_\ell(\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j)$, which with the noise gives (11.2). Each matrix $\sum_m Y_{\ell m}(\hat{\mathbf{n}}_i) Y_{\ell m}^*(\hat{\mathbf{n}}_j)$ is a Gram matrix and positive semidefinite; adding $\sigma_N^2 \mathsf{I}$ makes every eigenvalue at least $\sigma_N^2 > 0$.

Proposition B.2: Funk–Hecke smoothing

A bounded zonal profile $f(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}_0)$ smoothed by a zonal beam with Legendre coefficients T_ℓ is $\sum_\ell \frac{2\ell+1}{4\pi} f_\ell T_\ell P_\ell(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}_0)$, with $f_\ell = 2\pi \int_0^\pi f(\cos\theta) P_\ell(\cos\theta) \sin\theta\, d\theta$.

Proof. Expand profile and beam in Legendre polynomials, use the lemma to write each as a sum over m, and integrate the product over the sphere: orthonormality of the $Y_{\ell m}$ collapses the double sum, and the lemma once more returns a Legendre series with coefficients $f_\ell T_\ell$. □

Proof of Proposition 11.1. For $\mathbf{x} \neq \mathbf{0}$,

$$\mathbf{x}^{\top}\mathsf{A}\mathbf{x} = (\mathsf{T}^{\top}\mathbf{x})^{\top}\mathsf{C}^{-1}(\mathsf{T}^{\top}\mathbf{x}) > 0,$$

so A is positive definite. Expanding, $\chi^2(\mathbf{a}) = \chi^2(\mathbf{0}) - 2\mathbf{a}^{\top}\mathbf{b} + \mathbf{a}^{\top}\mathsf{A}\mathbf{a}$; completing the square with $\mathbf{b} = \mathsf{A}\hat{\mathbf{a}}$ gives the stated form. The estimate $\hat{\mathbf{a}} = \mathsf{A}^{-1}\mathsf{T}\mathsf{C}^{-1}\mathbf{d}$ is linear in $\mathbf{d}$; for $\mathbf{d} = \mathsf{T}^{\top}\mathbf{a} + \mathbf{n}$ its mean is $\mathbf{a}$ and its covariance is

$$\mathsf{A}^{-1}\mathsf{T}\mathsf{C}^{-1}\mathsf{C}\mathsf{C}^{-1}\mathsf{T}^{\top}\mathsf{A}^{-1} = \mathsf{A}^{-1}.$$

Under $\mathfrak{H}_0$, $\mathbf{b} \sim \mathcal{N}(\mathbf{0}, \mathsf{A})$; with $\mathsf{A} = \mathsf{R}^{\top}\mathsf{R}$ the vector $\mathbf{u} = \mathsf{R}^{-\top}\mathbf{b}$ is standard normal and $\Delta\chi^2 = \mathbf{u}^{\top}\mathbf{u} \sim \chi^2_k$.

Harmonic form. On the full sky with isotropic covariance, C^{-1} acts on harmonic coefficients as multiplication by $1/S_\ell$. A template centred at $\hat{\mathbf{n}}_0$ has, by Proposition B.2, coefficients $f_\ell T_\ell Y^*_{\ell m}(\hat{\mathbf{n}}_0)$. Hence $b_k = \sum_{\ell m} f_{k\ell} T_\ell S_\ell^{-1} d_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}_0)$ (the data being real), and

$$A_{kk'} = \sum_{\ell m} f_{k\ell} f_{k'\ell} T_\ell^2 S_\ell^{-1} |Y_{\ell m}(\hat{\mathbf{n}}_0)|^2,$$

which by the lemma at $\hat{\mathbf{n}} = \hat{\mathbf{n}}'$ is (13.1), independent of $\hat{\mathbf{n}}_0$.

Appendix C. The Poisson limit

If the number of detected imprints is Poisson with mean $\mu = \eta\bar{N}$, the probability of observing none is $e^{-\mu}$. The classical 95 per cent upper limit is the μ at which this probability is 0.05, namely $\mu = -\ln 0.05 = 2.996$; for k observed events the limit is $\tfrac{1}{2}\chi^2_{0.95}(2k+2)$, which reduces to the same number at $k = 0$. False positives only add to the observed count, so ignoring them makes (14.1) conservative. Because η increases with amplitude, the bound at a given amplitude holds for every imprint at least that bright.

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